Quantum Conflict Measurement in Decision Fusion for Out-of-Distribution Detection

Yilin Dong, Tianyun Zhu, Xinde Li, *Senior Member, IEEE*, Jean Dezert, Rigui Zhou, Changming Zhu, Lei Cao, and Shuzhi Sam Ge, *Fellow, IEEE*

***Abstract*—Quantum Dempster–Shafer theory (QDST) derives a quantum mass function (QMF), a fuzzy metric obtained from multiple information sources based on quantum interference. In general, QMF effectively represents and processes uncertain information, but managing conflicts among multiple QMFs remains challenging. To address this issue, we propose a novel quantum conflict indicator (QCI) within the QDST framework. It is the first metric satisfying ideal conflict measurement properties, including non-negativity, symmetry, boundedness, extreme consistency, and insensitivity to refinement. Based on QCI, a novel quantum conflict fusion method (QCI-Fusion) is introduced to fuse highly conflicting QMFs. Moreover, traditional methods, including QCI-Fusion, typically constructs the Quantum Frame of Discernment (QFoD) based on predicted labels, which makes it difficult to cover unseen classes. Therefore, a new decision architecture, QCI-Decision, is proposed for unsupervised detection that rejects out-of-distribution (OOD) samples while maintaining in-distribution (ID) classification. Experimental results show that the classification accuracy of QCI-Decision deviates from the original model predictions by at most 1.49%. Meanwhile, compared with the latest OOD detection methods, QCI-Decision improves the Area Under the Receiver Operating Characteristic Curve (AUC) by up to 0.6% and reduces the False Positive Rate at 95% True Negative Rate (FPR) by up to 1.63%. Moreover, compared with QCI-Fusion, QCI-Decision achieves approximately threefold faster fusion speed with negligible performance degradation, offering a promising solution for open-world quantum information decision.**

***Index Terms*—Quantum dempster-shafer theory, quantum mass function, quantum conflict indicator, conflict fusion method, out-of-distribution detection.**

Received 7 August 2024; revised 10 March 2026; accepted 24 April 2026. Date of publication 29 April 2026; date of current version 6 August 2026. This work was supported by the National Natural Science Foundation of China under Grant 62103258, Grant 62233003, and Grant 62073072. Recommended for acceptance by B. Rosenhahn. *(Corresponding authors: Xinde Li; Rigui Zhou.)*

Yilin Dong, Tianyun Zhu, Rigui Zhou, Changming Zhu, and Lei Cao are with the College of Information Engineering, Shanghai Maritime University, Shanghai 201306, China (e-mail: yldong@shmtu.edu.cn; 202330310212@stu.shmtu.edu.cn; rgzhou@shmtu.edu.cn; cmzhu@shmtu.edu.cn; lcao@shmtu.edu.cn).

Xinde Li is with the School of Automation, Southeast University, Nanjing 210096, China, and also with the Faculty of Robot Science and Engineering, Northeastern University, Shenyang 110004, China (e-mail: xindeli@seu.edu.cn).

Jean Dezert is with Information Processing and Systems Department, ONERA—The French Aerospace Lab, F-91123 Palaiseau, France (e-mail: jean.dezert@onera.fr).

Shuzhi Sam Ge is with the Social Robotics Laboratory, Department of Electrical and Computer Engineering, Interactive Digital Media Institute, National University of Singapore, Singapore 119077, and also with the Institute for Future, Qingdao University, Qingdao 266071, China (e-mail: samge@nus.edu.sg).

Digital Object Identifier 10.1109/TPAMI.2026.3688924

I. Introduction

In artificial intelligence systems, a fundamental challenge lies in representing and processing knowledge within uncertain environments to support decision-making. Numerous methods have been proposed to quantify and manage uncertainty, including fuzzy sets [1], rough sets [2], Z-numbers [3], Dempster-Shafer Theory of evidence [4], D-numbers [5], Entropy [6] and other hybrid methods [7]. These techniques have garnered significant attention across applications such as image classification [8], medical diagnosis [9], information fusion [10], and decision-making [11]. Among these approaches, quantum Dempster-Shafer theory (QDST) [12], a generalization of Dempster-Shafer theory (DST), is frequently employed to represent and manage uncertain information. Within QDST, a quantum mass function (QMF), also called a quantum basic belief assignment (QBBA), incorporates quantum probability and phase angle to respectively capture knowledge uncertainty and assess source reliability. Furthermore, QDST introduces the generalized Dempster rule of combination (DRC-QM) for uncertain reasoning, enabling it to retain the strengths of DST while enhancing its capability to handle uncertain information effectively.

However, QDST often struggles to handle conflicting information, which diminishes its decision-making effectiveness. Measuring the discrepancies among QMFs are essential to the effectiveness of the combination for multiple sources of evidence [13], [14]. Pan [12] introduced the conflict coefficient qK as a measurement of deviation among QMFs. Deng [15] proposed the distance $\mathbb{D}^p$ for QDST based on the Jousselme's distance. Meanwhile, Xiao [16] presented the complex Jensen–Shannon divergence CJS on the basis of Shannon entropy. After CJS divergence was proposed, $CBKL$ divergence [17], $CBJS$ divergence [17] and CB_{χ^2} divergence [18] were also given to consider the variation of phase angle. Although these methods can quantify discrepancy among QMFs in some cases, they may yield counter-intuitive or ineffective results. For instance, consider a numerical example:

Example 1.1: Assume that there exists three QMFs qm_1, qm_2 and qm_3 on $2^{|\Theta\rangle}_{refined}$ when the FoD is $|\Theta\rangle = \{|a_1\rangle, |a_2\rangle, |a_3\rangle\}$:

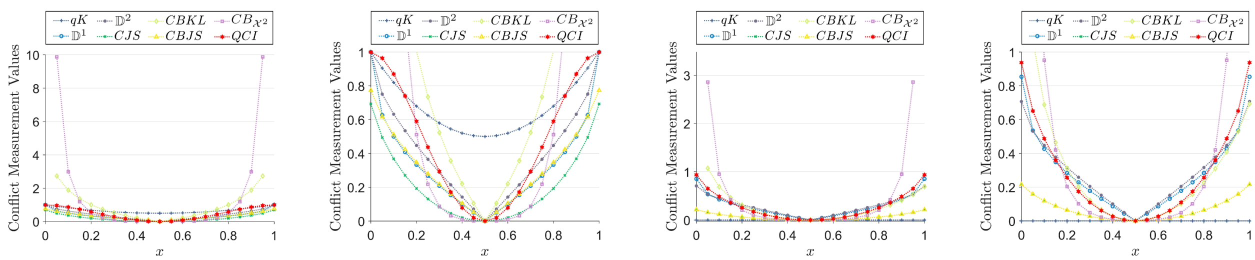
$$qm_1 : qm_1(|a_1\rangle) = \sqrt{0.8}, qm_1(|a_2\rangle) = \sqrt{0.2};$$

$$qm_2 : qm_2(|a_1\rangle) = \sqrt{0.8}e^{\mathrm{i}\frac{\pi}{2}}, qm_2(|a_2\rangle) = \sqrt{0.2}e^{\mathrm{i}\frac{\pi}{2}};$$

$$qm_3 : qm_3(|a_1\rangle) = \sqrt{0.7}, qm_3(|a_2\rangle) = \sqrt{0.2},$$

TABLE I
COMPARISONS OF CLASSICAL CONFLICT METHODS IN EXAMPLE 1.1

QMFs	qK	$\mathbb{D}^1$	$\mathbb{D}^2$	CJS	$CBKL$	$CBJS$	CB_{χ^2}
qm_1 and qm_2	0.32	0.7071	1	0	6.2832	3.4339	/
qm_1 and qm_3	0.4	0.1271	0.2273	0.0363	0.5008	0.2731	/
qm_2 and qm_3	0.4	0.7388	1	0.0363	5.9789	3.04	/



(a) When $\omega = |a_1\rangle$, the conflict measurement value adjustment caused by changes in x values.

(b) The amplification of (a) and the restriction of conflict measurement value within the interval $[0, 1]$.

(c) When $\omega = |a_1\rangle \cup |a_2\rangle$, the conflict measurement value adjustment caused by changes in x values.

(d) The amplification of (c) and the restriction of conflict measurement value within the interval $[0, 1]$.

Fig. 1. Comparisons of Sensitivity Curves for QCI and classical conflict methods in Example 3.1.

$$qm_3(|a_3\rangle) = \sqrt{0.1}e^{\mathrm{i}\frac{\pi}{2}}.$$

As shown in Table I, all the aforementioned conflict methods cannot distinguish the discrepancy among these three evidences as well. Specifically, the values of $qK(qm_1, qm_3)$ and $qK(qm_2, qm_3)$ are equal, and a similar trend is observed in CJS divergence. This is not intuitive because the phase angle of qm_1 is closer to that of qm_3 compared to qm_2. From the perspective of intuitiveness, the conflict between qm_2 and qm_3 should be greater than the conflict between qm_1 and qm_3. Meanwhile, the discrepancy between qm_2 and qm_3 should be greater than that between qm_1 and qm_2 because qm_1 and qm_2 share the same phase angles in $|a_1\rangle$ and $|a_2\rangle$, while qm_3 supports $|a_1\rangle$ less than qm_2 does. However, both $CBKL$ and $CBJS$ misevaluate the discrepancy information between them. Regarding $\mathbb{D}^p$, the choice of parameter p can lead to the completely contradictory results: $\mathbb{D}^1(qm_1, qm_2) < \mathbb{D}^1(qm_1, qm_3)$, while $\mathbb{D}^2(qm_1, qm_2) = \mathbb{D}^1(qm_1, qm_3)$. Furthermore, CB_{χ^2} divergence is unsuitable in this example. The issue illustrated in Example 1.1 indicates that the discrepancy information among QMFs extracted by all mainstream conflict methods in QDST is incomplete.

To address this problem, a more effective method for measuring discrepancy information among QMFs is required. In DST, several studies have identified two primary factors contributing to discrepancies among evidence elements: inconsistency and non-specificity [19], [20]. In QDST, the introduction of reliability further complicates these discrepancies, making traditional conflict methods inadequate for capturing them. A rigorous conflict method should be guided by two main ideas: 1) The natural extension of set theory: The conflict among QMFs should align with the natural conflict between sets. Specifically, when QMFs reduce to classical sets, conflict methods should correspond to the conflict among sets, ensuring logical consistency. 2) The principle of least commitment [21]: Conflict methods should not depend on predefined (in)dependence assumptions of information sources. Indeed, due to limited information, the dependency relationships among information sources are typically unknown. Consequently, conflict methods must be grounded in QMFs, without relying on unverified assumptions. Based on these principles, Destercke summarized five ideal attributes for conflict methods: non-negativity, symmetry, boundedness, extreme consistency, and insensitivity to refinement [22]. These properties ensure that the conflict measure behaves as a proper distance-like quantity in belief space and remains stable under different scenarios. However, existing conflict methods fall short. For instance, the lack of boundedness in $\mathbb{D}^p$ prevents a stable physical interpretation, making it impossible to set a common threshold for decision-making. This highlights the necessary for a new conflict method in QDST that incorporates these theoretical foundations.

The main contributions of this paper are as follows:

1) A novel conflict measurement in QDST, named Quantum Conflict Indicator (QCI), is proposed to effectively capture the discrepancy information among QMFs. QCI comprehensively accounts for the non-specificity, inconsistency and reliability of QMFs and is the first conflict measurement in QDST to embody all ideal conflict properties. Furthermore, based on QCI, a novel multi-source information fusion algorithm, named QCI-Fusion, is presented and several experiments have been proven that the given QCI-Fusion outperforms the state-of-the-art fusion methods.

2) Traditional QDST methods typically construct the QFoD based on predicted labels [23], [24]. As a result, they cannot cover unseen labels, which often leads to incorrect decisions and limits the applicability of QDST in open-world scenarios. To address this limitation, a decision architecture termed QCI-Decision is introduced for open-world unsupervised detection. Specifically, QCI-Decision constructs the QFoD using feature channels, where the feature information of each class is used to build feature centers. The conflict among QMFs—traditionally regarded as a discriminative burden—is transformed into a key

metric of semantic consistency. By explicitly quantifying the belief flow loss and orthonormalization between test samples and feature centers through QCI, QCI-Decision can effectively identify in-distribution (ID) samples while robustly rejecting out-of-distribution (OOD) samples. Experimental results show that the classification accuracy of QCI-Decision deviates from the original model predictions by at most 1.49%, with even smaller deviations for stronger models. Compared with the best baseline OOD detection methods, QCI-Decision improves AUC by up to 0.6% and reduces FPR by up to 1.63%. Meanwhile, the approximate computation and linear channel selection modules each accelerate runtime by about threefold with negligible performance loss, demonstrating the practicality of the proposed framework for open-world QDST scenarios.

The remainder of this paper is organized as follows: Section II introduces the background knowledge required for this paper. Section III presents the proposed QCI and its properties in detail. Numerical examples are also provided to compare QCI with other conflict measurements. Section IV presents QCI-Fusion and compares it with other fusion methods on the UCI dataset. Section V presents a new detection architecture, QCI-Decision and conducts a comprehensive evaluation using the Mini-ImageNet and CIFAR-10 datasets. Section VI concludes the paper.

II. Preliminaries

This section provides a brief introduction of QDST, its inherent concept of conflict and several classical conflict measurements. This background is essential for understanding the subsequent chapters of this paper.

A. Quantum Dempster-Shafer Theory

Assuming the QFoD $|\Theta\rangle = \{|a_1\rangle, \ldots, |a_n\rangle\}$ is an exclusive and non-empty set, whose elements are mutually exclusive. A QMF qm on the power set $2^{|\Theta\rangle} = \{\emptyset, A_1, \ldots, A_{2^n-1}\}$ can be defined as $qm(A_k) = \sqrt{\psi_k} \cdot e^{\mathrm{i}\theta_k}$, $k = 1, \ldots, 2^n - 1$, where i denotes the imaginary unit, which is defined by its property $\mathrm{i}^2 = -1$. Here, $qm(\emptyset) = 0$ and $\sum_{A_k \subseteq 2^{|\Theta\rangle}} |qm(A_k)|^2 = 1$ are satisfied. $qm(A_k)$ represents the amplitude given by Euler formula, which can be expressed as a complex number $x_k + y_k\mathrm{i}$. Specifically, $x_k = \sqrt{\psi_k}\cos(\theta_k)$ and $y_k = \sqrt{\psi_k}\sin(\theta_k)$. Moreover, ψ_k represents the quantum probability of A_k, defined as $\psi_k = |qm(A_k)|^2 = x_k^2 + y_k^2$, while θ_k represents the phase angle of A_k, ranging from the 0 to $\frac{\pi}{2}$. Furthermore, for any two QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}$, DRC-QM is defined as:

$$qm_1 \oplus qm_2(A_k) = \sqrt{\frac{(qm'(A_k))^2}{1 - qK}}, \tag{1}$$

$$qm'(A_k) = \begin{cases} \sum_{A_i \cap A_j = A_k} qm_1(A_i) \times qm_2(A_j), & A_k \neq \emptyset \\ 0, & A_k = \emptyset \end{cases} \tag{2}$$

where the conflict coefficient qK [12] is defined as:

$$qK = 1 - \sum_{A_k \subseteq 2^{|\Theta\rangle}} |qm'(A_k)|^2. \tag{3}$$

Note that if $qm(A_k) = \psi_k$, QDST degenerates to DST.

B. The Concept of Conflict Degree in QDST

A conflict between two QMFs can be interpreted qualitatively as one QMF strongly supports $A_i \subseteq 2^{|\Theta\rangle}$ and the other strongly supports $A_j \subseteq 2^{|\Theta\rangle}$, where A_i and A_j are not compatible [21]. Consider the following example:

Example 2.1: Let a QFoD be $|\Theta\rangle = \{|a_1\rangle, |a_2\rangle, |a_3\rangle\}$. Define two QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}_{refined}$ as follows

$$qm_1 : qm_1(|a_1\rangle) = \sqrt{\psi_1}e^{\mathrm{i}\theta_1}, qm_1(|a_2\rangle) = \sqrt{1-\psi_1};$$
$$qm_2 : qm_2(|a_3\rangle) = \sqrt{\psi_2}e^{\mathrm{i}\theta_2}, qm_2(|a_2\rangle) = \sqrt{1-\psi_2}.$$

Here, $\psi_1 = 0.9$ and $\psi_2 = 0.9$ are assigned. In this case, qm_1 strongly supports $|a_1\rangle$, and qm_2 strongly supports $|a_3\rangle$. By applying DRC-QM, the following results are obtained:

$$qm' : qm'(|a_2\rangle) = \sqrt{(1-0.9)(1-0.9)} = \sqrt{0.99},$$
$$qK = 1 - (1-0.9)(1-0.9) = 0.99,$$
$$qm_1 \oplus qm_2 : qm(|a_2\rangle) = 1.$$

Neither of the two strongly preferred choices by the two QMFs is preserved and the least preferred choice by the two QMFs is given the full quantum probability after the combination with DRC-QM. When this unreasonable decision happens, it is said that the two QMFs are in conflict.

Furthermore, the extreme case of maximal conflict appears when the supports of qm_1 and qm_2 are complete inconsistent [22], i.e., $\psi_1 = \psi_2 = 1$ and $\theta_1 = \theta_2$. In this case, qm_1 fully supports $|a_1\rangle$, and qm_2 fully supports $|a_3\rangle$. Conversely, the total absence of conflict occurs when qm_1 and qm_2 are complete consistent [22], i.e., $\psi_1 = \psi_2 = 0$. In this case, both qm_1 and qm_2 fully support $|a_2\rangle$ and no discrepancy exists between them.

C. Conflict Measurements of QDST

Definition 1: Conflict Coefficient qK: qK is defined in (3).

Definition 2: Distance of Deng $\mathbb{D}^p$ [15]: For any two QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}$, $\mathbb{D}^p$ is defined as follows:

$$\mathbb{D}^p(qm_1, qm_2) = \frac{\left[((Uqm_1 - Uqm_2)^{\frac{p}{2}})'((Uqm_1 - Uqm_2)^{\frac{p}{2}})\right]^{\frac{1}{p}}}{\left[(qm_1^{\frac{p}{2}})'(qm_1^{\frac{p}{2}}) + (qm_2^{\frac{p}{2}})'(qm_2^{\frac{p}{2}})\right]^{\frac{1}{p}}}, \tag{4}$$

where U represents the Cholesky decomposition of the Jaccard matrix between qm_1 and qm_2, and p is the integer greater than or equal to 1.

Definition 3: Complex Jensen–Shannon Divergence CJS [16]: For any two QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}$, CJS is defined as follows:

$$CJS(qm_1, qm_2) = \frac{1}{2}\sum_{k=1}^{n} N_{qm_1}(A_k) log\left(\frac{2N_{qm_1}(A_k)}{N_{qm_1}(A_k) + N_{qm_2}(A_k)}\right)$$

$$+\frac{1}{2}\sum_{k=1}^{n} N_{qm_2}(A_k) log\left(\frac{2N_{qm_2}(A_k)}{N_{qm_1}(A_k)+N_{qm_2}(A_k)}\right). \quad (5)$$

Here,

$$N(A_k)=\frac{|CBP(A_k)|}{\sum_{i=1}^{n}|CBP(A_i)|} \quad (6)$$

and

$$CBP(A_k)=\sum_{A_i\subseteq 2^{|\Theta\rangle}}|qm(A_i)|qm(A_i)\frac{|A_i\cap A_k|}{|A_i|}, \quad (7)$$

where $|A_i|$ represents the cardinality of A_i.

Definition 4: Complex Belief Kullback–Leibler Divergence $CBKL$ [17]: For any two QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}$, $CBKL$ is defined as follows:

$$CBKL(qm_1, qm_2) = \sum_{A_k\subseteq 2^{|\Theta\rangle}} CBP_{qm_1}(A_k)\log\left(\frac{CBP_{qm_1}(A_k)}{CBP_{qm_2}(A_k)}\right), \quad (8)$$

Definition 5: Complex Belief Jensen–Shannon Divergence $CBJS$ [17]: For any two QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}$, $CBJS$ is defined as follows:

$$CBJS(qm_1, qm_2) = \frac{1}{2}CBKL(qm_1, qm_{avg})+\frac{1}{2}CBKL(qm_2, qm_{avg}), \quad (9)$$

where $qm_{avg}(A_k)=\sqrt{\frac{|qm_1(A_k)|^2+|qm_2(A_k)|^2}{2}}e^{\mathrm{i}\frac{\theta}{2}}, A_k\subseteq 2^{|\Theta\rangle}$, and θ is the sum phase angle of $qm_1(A_k)$ and $qm_1(A_k)$.

Definition 6: Complex Belief $\mathcal{X}^2$ Divergence $CB_{\mathcal{X}^2}$ [18]: For any two QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}$, $CB_{\mathcal{X}^2}$ is defined as follows:

$$CB_{\mathcal{X}^2}(qm_1, qm_2) = \frac{1}{2}\mathcal{X}^2(CBP_{qm_1}, CBP_{qm_2}) + \frac{1}{2}\mathcal{X}^2(CBP_{qm_2}, CBP_{qm_1}). \quad (10)$$

Here,

$$\begin{aligned}\mathcal{X}^2(CBP_{qm_1}, CBP_{qm_2}) &= \sum_{k=1}^{n}\left|\frac{CBP_{qm_1}(A_k)-CBP_{qm_2}(A_k)}{\sqrt{CBP_{qm_2}(A_k)}}\right|^2 \\ &+2\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}\left|\frac{CBP_{qm_1}(A_i)-CBP_{qm_2}(A_i)}{\sqrt{CBP_{qm_2}(A_i)}}\right| \\ &\cdot\left|\frac{CBP_{qm_1}(A_j)-CBP_{qm_2}(A_j)}{\sqrt{CBP_{qm_2}(A_j)}}\right|\cdot cos(\theta), \end{aligned} \quad (11)$$

where θ is the angle between $\frac{CBP_{qm_1}(A_i)-CBP_{qm_2}(A_i)}{\sqrt{CBP_{qm_2}(A_i)}}$ and $\frac{CBP_{qm_1}(A_j)-CBP_{qm_2}(A_j)}{\sqrt{CBP_{qm_2}(A_j)}}$.

III. A Novel Quantum Correlation Coefficient and Quantum Conflict Indicator

This section presents the Quantum Conflict Indicator (QCI) as a quantitative measure for evaluating conflicts among QMFs. It follows the natural extension of set theory and the principle of least commitment, which require the indicator to satisfy five properties [22]: 1) Non-negativity: It defines QCI as a measure of logical distance, ensuring that the results have a clear physical interpretation. 2) Symmetry: It reflects the equality of information sources, meaning that the degree of conflict depends solely on the interaction among QMFs, independent of the input order. 3) Boundedness: By standardizing the interval to $[0, 1]$, QCI provides a unified benchmark for setting decision thresholds in various application contexts. 4) Extreme Consistency: It ensures that QCI can return to the "complete compatibility" or "complete contradiction" states of classical logic under boundary conditions, thereby preserving logical continuity. 5) Insensitivity to Refinement: It guarantees the robustness of QCI, maintaining consistent semantic-level conflict evaluations across different granularities or dimensions of the QFoD.

A. Definitions of QCC and QCI

This section introduces the Quantum Correlation Coefficient (QCC) to quantify the degree of consistency among QMFs, and the inverse of QCC, namely QCI, to evaluate conflicts among QMFs.

Definition 7: Quantum Correlation Coefficient (QCC): For any two QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}$. QCC is defined as:

$$\text{QCC}(qm_1, qm_2)=\left[\frac{|\langle qm_1|qm_2\rangle|}{||qm_1||\cdot||qm_2||}\right]^4, \quad (12)$$

where $\langle qm_1|qm_2\rangle$ represents the inner product of qm_1 and qm_2, which is defined as:

$$\langle qm_1|qm_2\rangle = \sum_{i=1}^{2^n-1}\sum_{j=1}^{2^n-1}\overline{qm_1(A_i)}qm_2(A_j)\mathcal{G}(\dot{\theta}_{i,j})\frac{|A_i\cap A_j|}{|A_i\cup A_j|}, \quad (13)$$

$$\mathcal{G}(\dot{\theta}_{i,j})=\frac{cos(\dot{\theta}_{i,j})+1}{2}. \quad (14)$$

Here, $\overline{qm_1(A_i)}$ represents the conjugate of $qm_1(A_i)$, $\dot{\theta}_{i,j}$ denotes the difference in phase angles between A_i and A_j, and $|\cdot|$ represents the cardinality of a set. Furthermore, $||qm_1||$ represents the paradigm of qm_1, which is defined as:

$$||qm_1|| = [|\langle qm_1|qm_1\rangle|]^{\frac{1}{2}} = \left[\left|\sum_{i=1}^{2^n}\sum_{j=1}^{2^n}\overline{qm_1(A_i)}qm_1(A_j)\mathcal{G}(A_i, A_j)\frac{|A_i\cap A_j|}{|A_i\cup A_j|}\right|\right]^{\frac{1}{2}}. \quad (15)$$

Note that a larger value of $\text{QCC}(qm_1, qm_2)$ indicates a higher consistency between the QMFs. Therefore, if $\text{QCC}(qm_1, qm_2) = 1$, it implies that qm_1 and qm_2 are completely consistent; whereas if $\text{QCC}(qm_1, qm_2) = 0$, qm_1 and

qm_2 are completely inconsistent. Moreover, when QDST degenerates to DST, QCC degenerates to the ECC proposed by Xiao in [20].

QCC has the following properties:

1) *Non-negativity:* $\mathrm{QCC}(qm_1, qm_2)$ is not negative.
2) *Symmetry:* $\mathrm{QCC}(qm_1, qm_2) = \mathrm{QCC}(qm_2, qm_1)$.
3) *Boundedness:* $0 \leq \mathrm{QCC}(qm_1, qm_2) \leq 1$.
4) *Nondegeneracy:* $\mathrm{QCC}(qm_1, qm_2) = 1$, iff $qm_1 = qm_2$.

Proof (*Non-negativity*): For any two QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}$, we have

$$\mathrm{QCC}(qm_1, qm_2) = \left[\frac{|\langle qm_1|qm_2\rangle|}{||qm_1||\,||qm_2||}\right]^4 \geq 0.$$

Clearly, $\mathrm{QCC}(qm_1, qm_2)$ cannot be negative.

End of proof

Proof (*Symmetry*): For any two QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}$, we have

$$\langle qm_1|qm_2\rangle = \sum_{i=1}^{2^n}\sum_{j=1}^{2^n}\overline{qm_1(A_i)}qm_2(A_j)\mathcal{G}(\dot{\theta}_{i,j})\frac{|A_i \cap A_j|}{|A_i \cup A_j|},$$

$$\overline{\langle qm_2|qm_1\rangle} = \sum_{i=1}^{2^n}\sum_{j=1}^{2^n}qm_2(A_i)\overline{qm_1(A_j)}\mathcal{G}(\dot{\theta}_{i,j})\frac{|A_i \cap A_j|}{|A_i \cup A_j|}.$$

It is clear to see that $\langle qm_1|qm_2\rangle = \overline{\langle qm_2|qm_1\rangle}$. Therefore, we have

$$\left[\frac{|\langle qm_1|qm_2\rangle|}{||qm_1|| \cdot ||qm_2||}\right]^4 = \left[\frac{|\overline{\langle qm_2|qm_1\rangle}|}{||qm_2|| \cdot ||qm_1||}\right]^4$$
$$= \left[\frac{|\langle qm_2|qm_1\rangle|}{||qm_2|| \cdot ||qm_1||}\right]^4.$$

End of proof

Proof (*Boundedness*): For this property, we need to prove the following two propositions separately:

(1) $\mathrm{QCC}(qm_1, qm_2) \geq 0$,

Proof: Clearly, $\mathrm{QCC}(qm_1, qm_2) \geq 0$ can be proved due to the non-negativity of QCC.

End of proof

(2) $\mathrm{QCC}(qm_1, qm_2) \leq 1$.

Proof: for any two QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}$, (12) can be rewritten in the following form:

$$\mathrm{QCC}(qm_1, qm_2) = \frac{|qm_1^\dagger V^\top V qm_2|^4}{|qm_1^\dagger V^\top V qm_1|^2 \cdot |qm_2^\dagger V^\top V qm_2|^2},$$

where $qm_1 = [qm_1(\emptyset), \ldots, qm_1(A_{2^n-1})]^\top$ and $qm_1^\dagger$ represents the conjugate transpose of qm_1, idem qm_2. Here, the matrix G represents a lower triangular matrix [20] and $V^T V$ denotes a symmetric positive definite matrix related to $\mathcal{G}(\dot{\theta}_{i,j})$ and $\frac{|A_i \cap A_j|}{|A_i \cup A_j|}$. Therefore, from the triangle inequality, we can derive:

$$|V(qm_1 + qm_2)|^2 \leq (|V qm_1| + |V qm_2|)^2. \tag{16}$$

Since there exists $|qm_1^\dagger V^\top V qm_2| = |qm_2^\dagger V^\top V qm_1|$ and $qm_1^\dagger V^\top V qm_1 = (V\overline{qm_1})^\top \cdot V qm_1 = |V qm_1|^2$, we have

$$\iff |(qm_1 + qm_2)^\dagger V^\top V (qm_1 + qm_2)| \leq |qm_1^\dagger V^\top V qm_1|$$
$$+ |qm_2^\dagger V^\top V qm_2| + 2\sqrt{|qm_1^\dagger V^\top V qm_1| \cdot |qm_2^\dagger V^\top V qm_2|}$$
$$\iff |qm_1^\dagger V^\top V qm_1| + |qm_2^\dagger V^\top V qm_2| + |qm_2^\dagger V^\top V qm_1|$$
$$+ |qm_1^\dagger V^\top V qm_2| \leq |qm_1^\dagger V^\top V qm_1| + |qm_2^\dagger V^\top V qm_2|$$
$$+ 2\sqrt{|qm_1^\dagger V^\top V qm_1| \cdot |qm_2^\dagger V^\top V qm_2|}$$
$$\iff \frac{|qm_1^\dagger V^\top V qm_2|^2}{|qm_1^\dagger V^\top V qm_1| \cdot |qm_2^\dagger V^\top V qm_2|} \leq 1$$

Clearly, $\mathrm{QCC}(qm_1, qm_2) \leq 1$ can be conducted.

End of proof

Proof: (*Nondegeneracy*): Based on (16), we can establish the following theorem: The condition for the equality sign in the triangular inequality is exclusively satisfied when the three points are collinear [26]. It is important to note that qm_1 and qm_2 are collinear only when qm_1 is equal to qm_2. As a result, the presence of $\mathrm{QCC}(qm_1, qm_2) = 1$ can only occur under the condition that $qm_1 = qm_2$.

End of proof

Definition 8: For any two QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}$, QCI is defined as:

$$\mathrm{QCI}(qm_1, qm_2) = 1 - \mathrm{QCC}(qm_1, qm_2). \tag{17}$$

Note that a larger value of $\mathrm{QCI}(qm_1, qm_2)$ indicates a higher conflict between the QMFs. Therefore, if $\mathrm{QCI}(qm_1, qm_2) = 1$, it implies a complete conflict between qm_1 and qm_2, which means two QMFs support completely different propositions; whereas if $\mathrm{QCI}(qm_1, qm_2) = 0$, qm_1 and qm_2 are completely non-conflicting, which means two QMFs support same propositions.

QCI has desirable properties required for a conflict measurement [27], including:

1) *Non-negativity:* $\mathrm{QCI}(qm_1, qm_2)$ is not negative.
2) *Symmetry:* $\mathrm{QCI}(qm_1, qm_2) = \mathrm{QCI}(qm_2, qm_1)$.
3) *Boundedness:* $0 \leq \mathrm{QCI}(qm_1, qm_2) \leq 1$.
4) *Extreme consistency:* 1) $\mathrm{QCI}(qm_1, qm_2) = 1$, iff for A_i and A_j of qm_1 and qm_2 respectively, $(\cup A_i) \cap (\cup A_j) = \emptyset$; 2) $\mathrm{QCI}(qm_1, qm_2) = 0$, iff $qm_1 = qm_2$.
5) *Insensitivity to refinement:* for qm_1 and qm_2 refined from $2^{|\Theta\rangle_1}$ to $2^{|\Theta\rangle_2}$, $\mathrm{QCI}(qm_1, qm_2)$ in $2^{|\Theta\rangle_1}$ is equal to $\mathrm{QCI}(qm_2, qm_1)$ in $2^{|\Theta\rangle_2}$.

Having established the properties of nondegeneracy, symmetry, and boundedness for QCC, it becomes evident that these properties are also satisfied for QCI. Furthermore, since the definition of QCI in this paper does not consider any aspects related to $2^{|\Theta\rangle}$, it follows that the proof of Insensitivity to refinement is of trivial nature [27]. Hence, the primary attention is directed towards investigating the property of Extreme consistency for QCI.

Proof: (*Extreme consistency*): To address this property adequately, it is necessary to provide separate proofs for the following two propositions:

1) $\mathrm{QCI}(qm_1, qm_2) = 1$, iff for A_i and A_j of qm_1 and qm_2 satisfy $(\cup A_i) \cap (\cup A_j) = \emptyset$, respectively.

TABLE II
THE RESULTS OF QCI IN EXAMPLE 1.1

QMFs	QCI
qm_1 and qm_2	0.9375
qm_2 and qm_3	0.1912
qm_3 and qm_4	0.9495

Proof: For two arbitrary QMFs qm_1 and qm_2 on $2^{|\Theta\rangle} = \{\emptyset, A_1, \ldots, A_{2^n-1}\}$, let us suppose that there exist non-zero elements $qm_1(A_1) \neq 0$ and $qm_2(A_1) \neq 0$ in qm_1 and qm_2, respectively. Additionally, it is assumed that there is no intersection between any other elements. Therefore, we have:

$$\mathrm{QCI}(qm_1, qm_2) = 1 - \left[\frac{|\langle qm_1|qm_2\rangle|}{||qm_1|| \cdot ||qm_2||}\right]^4.$$

Since both $||qm_1||$ and $||qm_2||$ are greater than zero and we have

$$\langle qm_1|qm_2\rangle = \overline{qm_1(A_1)}qm_2(A_1)\mathcal{G}(\dot{\theta_{1,1}}),$$

and

$$\mathrm{QCI}(qm_1, qm_2) = 1 - \left[\frac{|\overline{qm_1(A_1)}qm_2(A_1)\mathcal{G}(\dot{\theta_{1,1}})|}{||qm_1|| \cdot ||qm_2||}\right]^4 < 1.$$

Therefore, $\mathrm{QCI}(qm_1, qm_2) = 1$, iff for A_i and A_j of qm_1 and qm_2 respectively, $(\cup A_i) \cap (\cup A_j) = \emptyset$.

End of proof

2) $\mathrm{QCI}(qm_1, qm_2) = 0$, iff $qm_1 = qm_2$.

Proof: : By utilizing the Nondegeneracy of QCC, we can deduce that proposition $\mathrm{QCI}(qm_1, qm_2) = 0$ is valid iff $qm_1 = qm_2$.

End of proof

B. Physical Meaning of QCI

To further elucidate the internal mechanism of QCI, this section explores its physical interpretation. Unlike the real-valued mass function in DST, QMF is inherently a complex probability amplitude $qm(A_k) = \sqrt{\psi_k}e^{\mathrm{i}\theta_k}$, which is mathematically equivalent to the unit superposition state vector $\sum_k qm(A_k)|A_k\rangle$ in Hilbert space [28]. This representation enables the application of the quantum superposition principle to handle uncertain information. Within this framework, DRC-QM can be regarded as the coherent superposition and normalization of state vectors. Then, the conflict between QMFs can be physically characterized as follows: the orthonormalization of state vectors eliminates geometric overlap, resulting in a loss of belief flow during DRC-QM (as shown in Fig. 2). QCI converts the abstract loss of belief flow into a quantifiable value as follows: 1) $\overline{qm_1(A_i)}qm_2(A_j)$ in Eq (13) captures inconsistency information, quantifying the observed quantum probability loss. 2) $\frac{|A_i \cap A_j|}{|A_i \cup A_j|}$ in Eq (13) captures non-specificity information, quantifying the geometric overlap among state vectors. 3) $\mathcal{G}(\dot{\theta}_{i,j})$ in Eq (13) captures reliability information, quantifying the orthonormalization among state vectors. For clarity, we revisit Example 1.1.

The results of QCI are shown in Table II. Compared with the conflict methods in Table I, QCI provides more intuitive results, i.e., $\mathrm{QCI}(qm_1, qm_3) < \mathrm{QCI}(qm_1, qm_2)$ and $\mathrm{QCI}(qm_1, qm_2) < \mathrm{QCI}(qm_2, qm_3)$. The internal evolution of QCI in Example 1.1 can be explained as follows: First, for qm_1 and qm_3, $|a_3\rangle$ leads to local orthonormalization. However, since their quantum probabilities are highly consistent in $|a_1\rangle$ and $|a_2\rangle$, only a slight loss of belief flow occurs. Therefore, $\mathrm{QCI}(qm_1, qm_3)$ is minimized. Second, the quantum probabilities of qm_1 and qm_2 are identical, but the phase angle difference of $\frac{\pi}{2}$ results in $\mathcal{G} = 0.5$. This leads to a greater loss of belief flow than difference in quantum probabilities, making $\mathrm{QCI}(qm_1, qm_2)$ larger. Finally, while the quantum probabilities of qm_2 and qm_3 on $|a_1\rangle$ and $|a_2\rangle$ are highly consistent, the phase angle difference severely impacts the belief flow. Additionally, $|a_3\rangle$ results in $\frac{|A_i \cap A_j|}{|A_i \cup A_j|} = 0.67$. At this point, the belief flow experiences further loss, ultimately causing $\mathrm{QCI}(qm_2, qm_3)$ to reach its peak.

C. Numerical Simulation Experiments on QCI

This section presents three numerical examples to illustrate the advantages of QCI from the perspectives of inconsistency, non-specificity and reliability, respectively. Meanwhile, real-world motor rotor and Iris data are used to validate the advantages of QCI and to demonstrate the necessity of its properties in practical applications.

Example 3.1: This example demonstrates that QCI quantifies the more inconsistency information among QMFs than classical conflict methods. Let a QFoD be $|\Theta\rangle = \{|a_1\rangle, |a_2\rangle\}$, two QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}_{refined}$ are defined as follows:

$$qm_1 : qm_1(|a_1\rangle) = \sqrt{x}e^{\mathrm{i}\frac{\pi}{3}}, qm_1(\omega) = \sqrt{1-x}e^{\mathrm{i}\frac{\pi}{3}};$$
$$qm_2 : qm_2(|a_1\rangle) = \sqrt{1-x}e^{\mathrm{i}\frac{\pi}{3}}, qm_2(\omega) = \sqrt{x}e^{\mathrm{i}\frac{\pi}{3}}.$$

Here, $\omega = |a_1\rangle$ or $|a_1\rangle \cup |a_2\rangle$, and x is varied in the interval $[0, 1]$. When $\omega = |a_1\rangle$, as x varies in Fig. 1(a), all conflict measurements show the same trend of change. Specially, when $x = 0.5$, we have $qm_1(|a_1\rangle) = qm_2(|a_1\rangle)$ and $qm_1(|a_2\rangle) = qm_2(|a_2\rangle)$. At this point, qm_1 and qm_2 coincide exactly, whereas the value of qK is 0.5, which inaccurately estimates the discrepancy information. Conversely, when $x = 0$, we have $qm_1(|a_1\rangle) = qm_2(|a_2\rangle) = e^{\mathrm{i}\frac{\pi}{3}}$. Here, the discrepancy between qm_1 and qm_2 reaches its maximum. However, $CBKL$ and CB_{χ^2} are limited by inherent flaws in their definitions, i.e., the condition $CBP_{qm_1}(|a_1\rangle) = 0$ results in the inability to compute Eq (5) and Eq (10). Furthermore, the methods, D^p, CJS and $CBJS$, do not attain their maximum values at this point. Only QCI achieves its maximum value of 1, effectively capturing richer inconsistent information among QMFs and outperforming other methods. In addition, as shown in Fig. 1(b) and Fig. 1(d), the QCI curve changes more rapidly than other conflict measurements. This smoothness and rapid variation effectively reduce the ambiguous regions in the middle, thereby enhancing certainty and stability during conflict discrimination.

Example 3.2: In this case, we demonstrate that QCI quantifies the more non-specificity information among QMFs than classical conflict methods. Let the involved QFoD is $|\Theta\rangle = \{|a_1\rangle, |a_2\rangle, |a_3\rangle, |a_4\rangle\}$, four QMFs qm_1, qm_2, qm_3 and

TABLE III
COMPARISONS BETWEEN QCI AND CLASSICAL CONFLICT METHODS IN EXAMPLE 3.2 FROM THE PERSPECTIVE OF NON-SPECIFICITY

QMFs	qK	$\mathbb{D}^1$	$\mathbb{D}^2$	CJS	$CBKL$	$CBJS$	$CB_{\mathcal{X}^2}$	QCI
qm_1 and qm_2	0	0.4645	0.5412	0	0	0	/	0.75
qm_2 and qm_3	0	1.4194	0.7758	0.2253	1.8365	0.4970	/	0.9215
qm_3 and qm_4	0	0.7341	0.4351	0.0144	0.2103	0.0422	/	0.4375

TABLE IV
SATISFACTION OF THE IDEAL CONFLICT PROPERTIES BY QCI COMPARED TO OTHER METHODS

Properties	Non-negativity	Symmetry	Boundedness	Extreme consistency	Insensitivity to refinement
qK	✓	✓	✓	–	✓
$\mathbb{D}^p$	✓	✓	–	–	✓
CJS	✓	✓	✓	–	✓
$CBKL$	✓	–	–	–	–
$CBJS$	✓	✓	–	–	–
$CB_{\mathcal{X}^2}$	✓	✓	–	–	–
QCI	✓	✓	✓	✓	✓

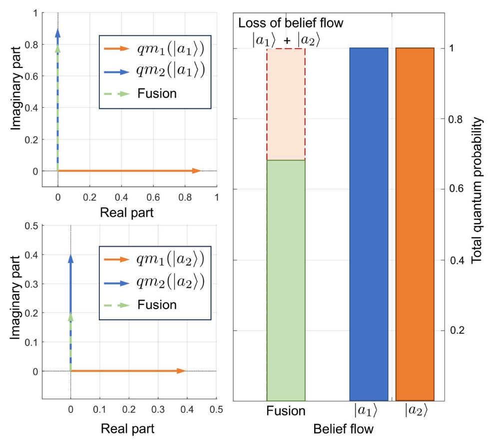


Fig. 2. Geometric manifestation of belief flow loss via Example 1.1: Orthonormalization overlap between qm_1 and qm_2.

qm_4 on $2^{|\Theta\rangle}_{refined}$ are defined as follows:

$$qm_1 : qm_1(|a_1\rangle) = \sqrt{0.5}e^{\mathrm{i}\frac{\pi}{3}}, qm_1(|a_2\rangle) = \sqrt{0.5}e^{\mathrm{i}\frac{\pi}{3}};$$
$$qm_2 : qm_2(|a_1\rangle \cup |a_2\rangle) = e^{\mathrm{i}\frac{\pi}{3}};$$
$$qm_3 : qm_3(|a_1\rangle \cup |a_2\rangle \cup |a_3\rangle) = \sqrt{0.5}e^{\mathrm{i}\frac{\pi}{3}},$$
$$qm_3(|a_2\rangle \cup |a_3\rangle \cup |a_4\rangle) = \sqrt{0.5}e^{\mathrm{i}\frac{\pi}{3}};$$
$$qm_4 : qm_4(|a_1\rangle \cup |a_2\rangle \cup |a_3\rangle \cup |a_4\rangle) = e^{\mathrm{i}\frac{\pi}{3}}.$$

As shown in Table III, all classical conflict methods cannot distinguish the discrepancy among qm_1, qm_2, qm_3 and qm_4. Specifically, the values of qK, CJS, $CBKL$ and $CBJS$ between qm_1 and qm_2 are 0. This is not intuitive because qm_1 supports $|a_1\rangle$ and $|a_2\rangle$, whereas qm_2 supports $|a_1\rangle \cup |a_2\rangle$. A discrepancy thus exists between qm_1 and qm_2, but the aforementioned methods fail to detect such non-specificity. Meanwhile, $\mathbb{D}^p$ is affected by the parameter p, and the extracted non-specific information may lead to contradictory results, such as $\mathbb{D}^1(qm_3, qm_4) > \mathbb{D}^1(qm_2, qm_2)$ and $\mathbb{D}^2(qm_3, qm_4) < \mathbb{D}^2(qm_2, qm_2)$. Furthermore, $CB_{\mathcal{X}^2}$ remains unavailable for this example. In contrast, QCI provides a more robust representation and more accurately captures the non-specific information among QMFs, yielding outcomes consistent with theoretical analysis.

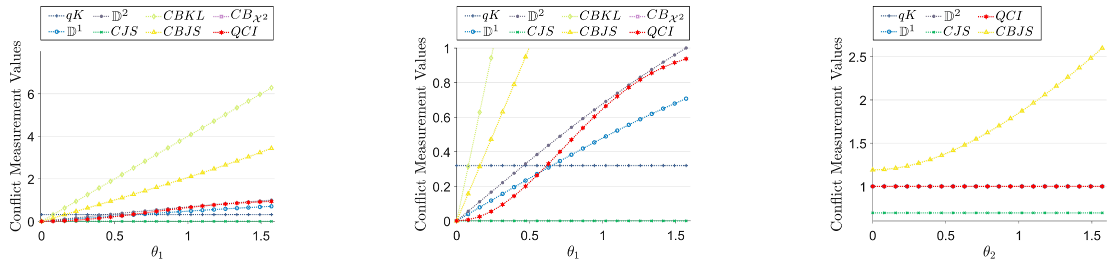
Example 3.3: Actually, the proposed QCI has the corresponding characteristic that such indicator is the more sensitive to the reliability information among QMFs than classical conflict methods. Let the QFoD is $|\Theta\rangle = \{|a_1\rangle, |a_2\rangle, |a_3\rangle\}$, three QMFs qm_1 and qm_2 on $2^{|\Theta\rangle}_{refined}$ are defined as follows:

$$qm_1 : qm_1(|a_1\rangle) = \sqrt{0.8}, qm_1(|a_2\rangle) = \sqrt{0.2};$$
$$qm_2 : qm_2(|a_1\rangle) = \sqrt{0.8}e^{\mathrm{i}\theta_1}, qm_2(|a_2\rangle) = \sqrt{0.2}e^{\mathrm{i}\theta_1};$$
$$qm_3 : qm_3(|a_3\rangle) = e^{\mathrm{i}\theta_2}.$$

Here, θ_1 and θ_2 are constrained to $[0, \frac{\pi}{2}]$. As show in Fig. 3(a) and (b), all conflict measurements exhibit a monotonically increasing trend with increasing θ_1, except for qK and CJS, which fail to capture any changes in reliability information. As show in Fig. 3(c), as θ_2 increases, $CBJS$ increases. This is not intuitive because regardless of the variation in θ_2, qm_1 only supports $|a_1\rangle$ and $|a_2\rangle$, whereas qm_3 supports $|a_3\rangle$. Their viewpoints are fundamentally conflicting. Meanwhile, $CBJS$ and $CB_{\mathcal{X}^2}$ cannot be reliably applied in this example. Compared to classical conflict methods, QCI not only accurately capture reliability information among QMFs but also produces a smoother curve with more rapid changes, demonstrating greater sensitivity and discriminative capability.

Example 3.4: This example presents a simple real application of the fault diagnosis for motor rotors [27]. Three types of sensors, positioned at different locations, collect acceleration, velocity and displacement data. The motor rotor has four states represented by $|\Theta\rangle$, i.e., $|a_1\rangle$: "normal operation", $|a_2\rangle$: "unbalanced", $|a_3\rangle$: "misaligned" and $|a_4\rangle$: "loose base". To highlight the necessity of QCI's properties in practical scenarios, only the data from the acceleration and velocity sensors are used to construct the QMFs as follows:

$$qm_1 : qm_1(|a_1\rangle) = \sqrt{0.06}, qm_1(|a_2\rangle) = \sqrt{0.68},$$
$$qm_1(|a_3\rangle) = \sqrt{0.02}, qm_1(|a_4\rangle) = \sqrt{0.04},$$



(a) The conflict measurement value adjustment between qm_1 and qm_2 caused by changes in θ_1 values.

(b) The amplification of (a) and the restriction of conflict measurement value within the interval $[0, 1]$.

(c) The conflict measurement value adjustment between qm_1 and qm_3 caused by changes in θ_2 values.

Fig. 3. Comparisons of Sensitivity Curves for QCI and classical conflict methods in Example 3.3.

TABLE V
THE FAULT DIAGNOSIS CASE FOR COMPARISONS BETWEEN THE PROPOSED QCI AND OTHER METHODS

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
qK	0.669	0.669	0.669	0.669
$\mathbb{D}^1$	0.5441	0.6799	0.8555	1.0109
$\mathbb{D}^2$	0.794	0.8092	0.8494	0.9014
CJS	0.3862	0.3882	0.3948	0.4071
$CBKL$	6.5231	6.2892	5.6461	4.7808
$CBJS$	1.1231	1.1568	1.2156	1.2538
CB_{χ^2}	6.6408	6.7678	7.2561	8.4391
QCI	0.9346	0.9458	0.9702	0.9884

$$qm_1(|\Theta\rangle) = \sqrt{0.2}e^{\mathrm{i}\theta};$$

$$qm_2 : qm_2(|a_1\rangle) = \sqrt{0.02}, qm_2(|a_3\rangle) = \sqrt{0.79},$$

$$qm_2(|a_4\rangle) = \sqrt{0.05}, qm_2(|\Theta\rangle) = \sqrt{0.14}.$$

As shown in Table IV, due to the boundedness property, the qK, QCI and CJS methods can determine whether two pieces of evidence conflict by setting a conflict detection threshold. However, when the threshold is set to 0.7, only QCI identifies qm_1 and qm_2 as conflicting in Table V. This indicates that QCI is more sensitive to the discrepancy among QMFs, allowing it to detect more inconsistency and non-specificity information among them. Furthermore, since conflict measurements quantify the degree of discrepancy among QMFs, their values must always be non-negative. This property is satisfied by all conflict methods. In addition, to ensure consistency in decision-making, the discrepancy between qm_1 and qm_2 should be equal to that value between qm_2 and qm_1. However, $CBKL$ does not satisfy this property. Secondly, boundedness and extreme consistency constrain the range and limiting behavior of the conflict methods. When information is expressed as a single set, the conflict measurements should attain its minimum value for complete consistency and its maximum value for complete inconsistency [29]. Failure to satisfy boundedness, such as $\mathbb{D}^p$, $CBKL$, $CBJS$ and CB_{χ^2}, results in the absence of well-defined limits for total consistency and inconsistency. Finally, insensitivity to refinement refers to the stability of the conflict measurements when $|\Theta\rangle$ is extended. For example, when new state $|a_5\rangle$ is added to the motor rotor, the discrepancy information between qm_1 and qm_2 should remain stable regardless of changes in $|\Theta\rangle$. However, $CBKL$, $CBJS$ and CB_{χ^2} do not satisfy this property. In conclusion, QCI meets all the ideal conflict properties, demonstrating superiority over classical conflict methods.

TABLE VI
QUANTUM MASS FUNCTIONS FOR EXAMPLE 3.5

Elements	qm_1	qm_2
$\lvert a_1\rangle$	$0.2750 - 0.0206\mathrm{i}$	$0.3329 - 0.0113\mathrm{i}$
$\lvert a_2\rangle$	$0.3925 - 0.0057\mathrm{i}$	$0.3825 - 0.0040\mathrm{i}$
$\lvert a_3\rangle$	$0.3932 + 0.0056\mathrm{i}$	$0.3862 + 0.0020\mathrm{i}$
$\lvert a_1\rangle \cup \lvert a_2\rangle$	$0.3912 - 0.0088\mathrm{i}$	$0.3838 - 0.0043\mathrm{i}$
$\lvert a_1\rangle \cup \lvert a_3\rangle$	$0.3918 - 0.0060\mathrm{i}$	$0.3854 - 0.0028\mathrm{i}$
$\lvert a_2\rangle \cup \lvert a_3\rangle$	$0.3926 - 0.0000\mathrm{i}$	$0.3856 - 0.0005\mathrm{i}$
$\lvert \Theta\rangle$	$0.3918 - 0.0060\mathrm{i}$	$0.3860 - 0.0017\mathrm{i}$

Example 3.5: This example uses QMFs derived from the Iris dataset [15] to further evaluate the performance of QCI when dealing with information sources that exhibit similar opinions. The original data, collected from four features (sepal length, sepal width, petal length, and petal width), is mapped into $2^{|\Theta\rangle}$, where $|\Theta\rangle = \{|a_1\rangle, |a_2\rangle, |a_3\rangle\}$ represents the three decision classes: setosa, versicolor, and virginica. For clarity, only the data from petal length and petal width (both supporting $|a_3\rangle$) are used, as shown in Table VI. The results are as follows: $qK = 0.2217$, $\mathbb{D}^1 = 0.022$, $\mathbb{D}^2 = 0.0466$, $CJS = 0.0001$, $CBKL = 0.0808$, $CBJS = 0.0417$, $CB_{\chi^2} = 0.0006$ and $\text{QCI} = 0.0023$. Since $\mathbb{D}^p$, $CBKL$, $CBJS$, and CB_{χ^2} do not satisfy the boundedness, it is difficult to objectively evaluate the degree of conflict. Meanwhile, qK produces a relatively high value, exaggerating the discrepancy between qm_1 and qm_2, whereas CJS yields a value close to zero, failing to distinguish between qm_1 and qm_2. However, the obtained value $\text{QCI} = 0.0023$ remains small enough to reflect the high consistency between qm_1 and qm_2, while still being non-zero, thereby preserving the ability to detect subtle discrepancies. This result further demonstrates the robustness of QCI compared with other conflict methods in real-world scenarios.

IV. QCI-BASED CONFLICT FUSION METHOD AND ITS APPLICATION IN PATTERN CLASSIFICATION

In multi-source information fusion systems, effectively managing conflicts among evidence is a core challenge in ensuring rational decision-making. The qualitative analysis in Section II-B indicates that significant conflict between QMFs may lead to decision failure in DRC-QM. Section III-B further clarifies the underlying cause of this phenomenon, demonstrating that the conflict between QMFs arises from the orthonormalization of state vectors. The absence of geometric overlap leads to a loss of belief flow during the fusion process. To address this loss caused by conflicts and reduce unreasonable decisions, this section proposes a novel multi-source information fusion method based on QCI, termed QCI-Fusion. Notably, to tackle the exponential increase in DRC-QM running time caused by the growing number of QMFs [15], we develop an approximate calculation module for QCI-Fusion. This module significantly reduces computational cost while maintaining fusion accuracy, making QCI-Fusion a promising solution for applications with limited computing power or high real-time requirements.

A. Method Describing

For a set QM with m QMFs in $2^{|\Theta\rangle} = \{\emptyset, A_1, \ldots, A_{2^n-1}\}$, the specific steps of the QCI-Fusion are as follows (step1-5 is the approximate calculation module):

step 1: Calculate the QCI between any two QMFs $qm_i, i = 1, \ldots, m$ and $qm_j, j = 1, \ldots, m$ by Eq (17) to generate the QCI matrix $\mathcal{Q}$ as follows:

$$\mathcal{Q} = \begin{bmatrix} \mathrm{QCI}(qm_1, qm_1) & \cdots & \mathrm{QCI}(qm_1, qm_m) \\ \vdots & \ddots & \vdots \\ \mathrm{QCI}(qm_m, qm_1) & \cdots & \mathrm{QCI}(qm_m, qm_m) \end{bmatrix}. \quad (18)$$

step 2: Define a conflict detection threshold λ_{cof} and filter $\mathcal{Q}$ to obtain $\mathcal{Q}'$ as follows:

$$\mathrm{QCI}(qm_i, qm_j) = \begin{cases} 0, \mathrm{QCI}(qm_i, qm_j) > \lambda_{cof} \\ \mathrm{QCI}(qm_i, qm_j), else \end{cases} \quad (19)$$

step 3: For each column in $\mathcal{Q}'$, count the number of non-zero elements and identify the index $Idx1$ of the column with the maximum count.

step 4: Define the subset of QMFs indexed by the non-zero elements in the $Idx1$-th column of $\mathcal{Q}'$ as a cluster. Select qm_{Idx1} as the representative QMF for this cluster and then remove all QMFs within this cluster from QM.

step 5: Repeat steps 1-4 for the remaining QMFs until no more clusters can be formed, i.e., when the maximum count of non-zero elements in any column of $\mathcal{Q}'$ is 1. The set of representative and remaining QMFs is denoted as QM'.

step 6: For each $qm_i, i = 1, \ldots, m'$ in QM', calculate its support degree as follows:

$$Sup_{qm_i} = \sum_{j=1, j\neq i}^{m'} (1 - \mathrm{QCI}(qm_i, qm_j)). \quad (20)$$

step 7: For each $qm_i, i = 1, \ldots, m'$ in QM', calculate its discount coefficient as follows:

$$D_{qm_i} = \sqrt{\frac{Sup_{qm_i}}{\sum_{j=1}^{m'} Sup_{qm_j}}}. \quad (21)$$

step 8: For each $qm_i, i = 1, \ldots, m'$ in QM', calculate its discount QMF as follows:

$$DMF_i = \begin{cases} D_{qm_i} \times qm_i(A_k), A_k \subseteq 2^{|\Theta\rangle}, A_k \neq |\Theta\rangle \\ \sum_{j=1}^{m'} (1 - D_{qm_i}) \times qm_i(A_k) + qm_i(|\Theta\rangle), else \end{cases} \quad (22)$$

step 9: After normalization of all discount QMFs, the discount QMFs are fused by (1) as follows:

$$qm_{fin} = ((DMF_1 \oplus DMF_2) \oplus \ldots) \oplus DMF_{m'}. \quad (23)$$

B. Application in Pattern Classification

To verify the performance of the QCI-Fusion method in open-world complex decision-making scenarios, this section applies it to the field of pattern classification tasks. Pattern classification seeks to classify objects of interest into one of several categories [30], and multi-source information fusion has been shown to be an effective method for improving classification performance in the presence of noise and conflicting interference [31]. The following experiments demonstrate how QCI-Fusion captures information loss caused by conflicts, corrects unreasonable decisions and achieves more accurate recognition than traditional methods in specific classification tasks.

1) Experiment Settings: Datasets: QCI-Fusion is applied to various commonly used datasets from the UC Irvine Machine Learning Repository (UCI),[1] including: 1) Iris dataset contains three classes: setosa, versicolor and virginica, comprising 150 samples. Each sample has four attributes: sepal length, sepal width, petal length, and petal width. 2) Banknote dataset contains two classes: genuine and forged banknotes, comprising 1372 samples. Each sample has four attributes extracted from sample images using the wavelet transform. 3) Transfusion dataset contains two classes: donating blood or not, comprising 178 samples. Each sample has four attributes: recency, frequency, monetary and time. 4) Seeds dataset has three classes: Kama, Rosa and Canadian wheat, comprising 210 samples. Each instance has seven attributes extracted from real samples using a soft X-ray technique. 5) Wholesale dataset has three classes, Lisbon, Oporto and Other Region, for a total of 440 samples. Each sample has seven attributes: channel, fresh, milk, grocery, frozen, detergents paper and delicassen. 6) Breast Cancer dataset has two classes, cancerous or non-cancerous, comprising 440 samples. Each sample has nine attributes: age, menopause, tumor-size, inv-nodes, node-caps, deg-malig, breast, breast-quad and irradiat.

Compared methods: QCI-Fusion is evaluated by comparing it against several representative fusion methods, including: 1) DRC [4] assumes that evidence is independent and resolves conflicts through a global normalization factor. 2) Murphy's method [32] averages multiple belief functions before applying DRC to ensure convergence in conflict scenarios. 3) The improved Proportional Conflict Redistribution Rule no.5 (PCR5+) [33] transfers the conflicting mass to all elements involved in this conflict and proportionally to their individual masses. 4) The improved Proportional Conflict Redistribution Rule no.6 (PCR6+) [33] no longer allocates the entire conflicting mass but instead uses the sum of their masses for transfer. 5) DRC-QM [12] extends mass function to the complex Hilbert space and uses quantum interference to simulate information interaction. 6) Deng's method [15] extends the Jousselme distance

[1] http://archive.ics.uci.edu/ml/index.php

TABLE VII
THE PERFORMANCE OF RECOGNITION IN SELECTED PERCENTAGES OF IRIS DATASET

Methods	10%		30%		50%		70%		90%		Average		
	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Running time↓
DRC	93.07	93.03	93.97	93.97	94.58	94.57	94.22	94.21	95.32	95.3	94.23	94.22	0.026
Murphy	93.11	93.07	94.01	94.01	94.67	94.65	94.38	94.37	95.26	95.24	94.29	94.27	0.009
PCR5+	93.36	92.96	94.46	94.45	94.99	94.96	94.65	94.64	95.46	95.45	94.58	94.49	0.051
PCR6+	93.36	93.32	94.44	94.44	94.97	94.94	94.6	94.59	95.46	95.45	94.57	94.55	0.061
DRC-QM	93.46	93.42	94.44	94.44	94.98	94.95	94.61	94.6	**95.79**	**95.78**	94.66	94.64	0.027
Deng p=1	93.46	93.41	**94.66**	**94.66**	95.17	95.14	94.78	94.77	95.67	95.66	94.75	94.73	0.046
Deng p=2	93.46	93.42	**94.66**	**94.66**	95.16	95.13	94.78	94.77	95.67	95.66	94.75	94.73	0.046
QCI-Fusion $\lambda_{cof} = 0$	**94.54**	**94.51**	94.65	94.65	**95.24**	**95.24**	**94.86**	**94.86**	95.74	95.74	**95.01**	**95**	0.046
QCI-Fusion $\lambda_{cof} = 0.25$	94.19	94.16	94.46	94.46	94.83	94.79	94.63	94.62	95.74	95.73	94.77	94.75	0.041 (0.005↓)
QCI-Fusion $\lambda_{cof} = 0.5$	94.19	94.16	94.46	94.46	94.83	94.79	94.63	94.62	95.74	95.73	94.77	94.75	0.041 (0.005↓)
QCI-Fusion $\lambda_{cof} = 0.75$	93.53	93.48	93.5	93.51	93.96	93.92	94.17	94.18	96.15	96.36	94.26	94.29	0.035 (0.011↓)

TABLE VIII
THE PERFORMANCE OF RECOGNITION IN SELECTED PERCENTAGES OF BANKNOTE DATASET

Methods	10%		30%		50%		70%		90%		Average		
	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Running time↓
DRC	86.95	83.28	85.05	78.23	85.48	79.14	86.63	81.32	84.7	73.98	85.76	79.19	0.0012
Murphy	87.09	86.19	85.05	84.19	85.61	84.56	86.78	85.96	84.72	83.46	85.85	84.87	0.0004
PCR5+	87.11	86.19	85.05	84.18	**86.98**	**85.9**	86.68	85.86	85.52	84.23	86.27	85.27	0.002
PCR6+	87.1	86.19	85.05	84.18	**86.98**	**85.9**	86.68	85.86	85.52	84.23	86.27	85.27	0.0017
DRC-QM	87.1	86.17	85.05	84.16	85.62	84.57	86.79	85.97	84.72	83.46	85.86	84.87	0.0013
Deng p=1	87.6	86.69	85.55	84.67	86.12	85.06	87.29	86.46	85.23	83.96	86.36	85.37	0.0035
Deng p=2	87.61	86.7	85.55	84.67	86.12	85.06	87.29	86.46	85.23	83.96	86.36	85.37	0.0035
QCI-Fusion $\lambda_{cof} = 0$	**88.17**	**87.44**	**86.18**	**85.44**	86.23	85.33	**87.46**	**86.74**	**86.02**	**84.83**	**86.81**	**85.96**	0.0034
QCI-Fusion $\lambda_{cof} = 0.25$	87.61	86.7	85.92	85.04	86.14	85.08	87.27	86.44	85.55	84.28	86.5	85.51	0.0032 (0.0002↓)
QCI-Fusion $\lambda_{cof} = 0.5$	87.61	86.7	85.92	85.04	86.14	85.08	87.27	86.44	85.55	84.28	86.5	85.51	0.0032 (0.0002↓)
QCI-Fusion $\lambda_{cof} = 0.75$	86.96	86.09	84.91	84.01	85.08	84.04	86.03	85.13	84.87	83.58	85.57	84.57	0.0026 (0.0007↓)

to QDST for conflict measurement and performs multi-source information fusion based on a discount coefficient.

Evaluation metrics: Three evaluation metrics are used to assess the performance of QCI-Fusion and other fusion methods, as follows: 1) Acc (%): Accuracy represents the ratio of correctly predicted samples to the total number of samples, which reflects the overall effectiveness of fusion methods in pattern classification. Higher values indicate better performance. 2) F1 (%): F1-score considers both precision and recall of the classification results, which ensures a robust evaluation of the fusion methods. Higher values indicate better performance. 3) Running time (s): Running time, defined as the time from the input of original QMFs to the decision output, is used to evaluate the practical applicability of the fusion method in time-sensitive or resource-constrained environments. Lower values indicate better performance.

Experiment settings: The experimental procedure presented by Deng in [15] is selected as follows:

step 1: For a dataset with n classes and m attributes, the training set and test set are selected from the dataset. Here, the dataset is one of the following: Iris dataset, Banknote dataset, Transfusion dataset, Seeds dataset, Wholesale dataset and Breast Cancer dataset.

step 2: The QFoD of the dataset is denoted as $|\Theta\rangle = \{|a_1\rangle, \ldots, |a_n\rangle\}$, where $\{|a_k\rangle \in |\Theta\rangle$ represents the k-th class of the dataset. For each attribute $i, i = 1, \ldots, m$, the mean μ_i^k and standard deviation σ_i^k of the training set for $A_k \subseteq 2^{|\Theta\rangle}$ are calculated.

step 3: An unknown target from the test set is selected and added to the training set. For each attribute $i, i = 1, \ldots, m$, the mean $\mu_i^{k'}$ and standard deviation $\sigma_i^{k'}$ of $A_k \subseteq 2^{|\Theta\rangle}$ are calculated again.

step 4: The original QMF of $A_k \subseteq 2^{|\Theta\rangle}$ for attribute $i, i = 1, \ldots, m$ is generated as follow:

$$qm_i(A_k) = \left(\frac{1}{e^{|\sigma_i^{k'} - \sigma_i^k|}} e^{\mathrm{i}|\mu_i^{k'} - \mu_j^k|} \right). \tag{24}$$

step 5: Normalize qm_i as follow:

$$qm_i(A_k) = \sqrt{\frac{(qm_i(A_k))^2}{\sum_{j=1}^{2^n-1} |qm_i(A_k)|^2}}. \tag{25}$$

step 6: The fusion of all qm_i is performed using the following methods respectively: QCI-Fusion, DRC, DRC-QM, the method of Deng, the method of Murphy, PCR5+ and PCR6+.

$$qm = ((qm_1 \oplus qm_2) \oplus \ldots) \oplus qm_m. \tag{26}$$

step 7: qm is converted into probability distribution using the quantum pignistic probability transform [15] for decision making.

$$QBetP(A_k) = \sum_{A_j \subseteq 2^{|\Theta\rangle}} \frac{|qm(A_j)|^2}{1 - |qm(\emptyset)|^2} \cdot \frac{|A_j \cap A_k|}{|A_j|}. \tag{27}$$

Different percentages of the dataset is randomly selected to serve as the training set, while the remaining data is used as the test set. For each scenario, one hundred Monte Carlo validation experiments are performed and the average results are reported.

2) Analysis of Results: Table VII–XII present the average accuracy, F1-score and running time of target recognition on the Iris, Banknote, Transfusion, Seeds, Wholesale and Breast Cancer datasets, respectively. For a comprehensive evaluation, the training set proportions are set to 10%, 30%, 50%, 70% and 90%. The highest accuracy and F1-score are highlighted in bold, and the second-highest values are underlined. Furthermore, for

TABLE IX
THE PERFORMANCE OF RECOGNITION IN SELECTED PERCENTAGES OF TRANSFUSION DATASET

Methods	10%		30%		50%		70%		90%		Average		
	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Running time↓
DRC	70.87	42.78	75.81	44.94	72.9	44.7	69.76	42.15	74.13	64.88	72.69	47.89	0.0013
Murphy	70.87	43.34	75.83	46.44	72.91	48.95	69.81	42.35	74.17	66.26	72.72	49.47	0.0004
PCR5+	70.86	43.34	75.75	46.07	73.55	48.69	71.1	42.5	73.29	65.22	72.91	49.16	0.0026
PCR6+	70.84	43.32	75.77	46.39	72.84	48.45	70.41	42.71	74.05	66.15	72.78	49.41	0.003
DRC-QM	70.8	43.26	76.55	47.21	72.90	49.26	69.65	42.93	74.77	66.96	72.93	49.92	0.0013
Deng p=1	71.49	43.69	76.77	49.84	71.65	47.91	70.6	43.52	74.97	66.98	73.1	50.39	0.0037
Deng p=2	69.16	**45.88**	64.57	**59.46**	59.7	**55.25**	62.19	**52.23**	69.07	55.9	64.94	**53.74**	0.0037
QCI-Fusion $\lambda_{cof}=0$	**72.74**	44.15	**76.99**	47.59	**74.49**	49.01	**71.29**	43.25	**75.51**	**67.46**	**74.2**	50.29	0.0036
QCI-Fusion $\lambda_{cof}=0.25$	72.59	44.87	**76.99**	47.59	**74.49**	49.01	**71.29**	43.25	**75.51**	**67.46**	74.17	50.44	0.0034 (0.0002↓)
QCI-Fusion $\lambda_{cof}=0.5$	72.59	44.87	**76.99**	47.59	**74.49**	49.01	**71.29**	43.25	**75.51**	**67.46**	74.17	50.44	0.0034 (0.0002↓)
QCI-Fusion $\lambda_{cof}=0.75$	72.51	44.83	**76.99**	47.59	**74.49**	49.01	**71.29**	43.25	**75.51**	**67.46**	74.16	50.43	0.0034 (0.0002↓)

TABLE X
THE PERFORMANCE OF RECOGNITION IN SELECTED PERCENTAGES OF SEEDS DATASET

Methods	10%		30%		50%		70%		90%		Average		
	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Running time↓
DRC	87.6	87.56	89.21	89.14	90.24	90.37	89.77	89.55	90.23	88.17	89.41	88.96	0.055
Murphy	87.29	87.11	88.75	88.85	89.92	90.02	89.65	89.62	90.13	90.1	89.15	89.14	0.009
PCR5+	**89.31**	**89.09**	**89.8**	**89.89**	89.86	89.92	90.48	90.37	90.48	90.45	**89.99**	**89.94**	80.195
PCR6+	88	87.77	88.78	88.88	90.34	90.42	89.68	89.62	90.48	90.45	89.46	89.43	74.215
DRC-QM	87.95	87.76	89.65	89.67	90.61	90.63	90.12	90.01	90.67	90.64	89.8	89.74	0.055
Deng p=1	87.7	87.51	89.45	89.52	80.48	90.54	90.4	90.29	90.58	90.55	89.72	89.68	0.117
Deng p=2	87.7	87.51	89.5	89.57	90.5	90.56	90.38	90.27	90.58	90.55	89.73	89.69	0.117
QCI-Fusion $\lambda_{cof}=0$	88.14	87.95	89.58	89.65	**90.65**	**90.71**	**90.56**	**90.45**	**90.75**	**90.72**	89.94	89.9	0.115
QCI-Fusion $\lambda_{cof}=0.25$	87.87	87.72	89.24	89.3	**90.65**	**90.71**	89.76	89.58	**90.75**	**90.72**	89.65	89.61	0.103 (0.012↓)
QCI-Fusion $\lambda_{cof}=0.5$	87.87	87.72	89.24	89.3	**90.65**	**90.71**	89.76	89.58	**90.75**	**90.72**	89.65	89.61	0.103 (0.012↓)
QCI-Fusion $\lambda_{cof}=0.75$	87.87	87.72	89.24	89.3	**90.65**	**90.71**	89.76	89.58	**90.75**	**90.72**	89.65	89.61	0.102 (0.013↓)

QCI-Fusion, we set λ_{cof} to 1 (in which case the approximate calculation module is disabled), as well as to 0.75, 0.5 and 0.25. The changes in running time are also reported.

The experimental results indicate that the average accuracy and F1-score of QCI-Fusion ($\lambda_{cof}=0$) are consistently higher than those of the other fusion methods across all datasets. For Iris dataset (Table VII), QCI-Fusion achieves the highest average accuracy (95.01%) and F1-score (95%), whereas the next-best method attains only 94.75% accuracy and 94.73% F1-score. For Banknote dataset (Table VIII), QCI-Fusion again yields the highest average accuracy (86.81%) and F1-score (85.96%), compared with the next-best method of 86.36% accuracy and 85.27% F1-score. For Transfusion dataset (Table IX), although Deng method's ($p=2$) achieves a higher F1-score (53.74%) than QCI-Fusion (50.29%), its average accuracy (64.94%) is substantially lower than that of the other methods (Deng method's ($p=1$)'s accuracy is 72.69%). Meanwhile, Deng's method ($p=1$) attains the highest accuracy among the methods (73.1%) except QCI-Fusion, indicating that the selection of parameter p significantly affects Deng method's performance and leads to instability in pattern classification. For Seeds dataset (Table X), the average accuracy (89.94%) and F1-score (89.9%) of QCI-Fusion are only slightly lower than those of PCR5+ (accuracy 89.99%, F1-score 89.94%). However, the running time of PCR5+ is approximately 700 times that of QCI-Fusion, and both PCR5+ and PCR6+ involve highly complex algorithms. Specifically, suppose there are m QMFs defined on $2^{|\Theta\rangle}$, containing n elements. The computational complexity of fusion methods is: PCR5+: $\mathrm{O}(\mathrm{m}^3 \times \mathrm{n}^{\mathrm{m}})$; PCR6+: $\mathrm{O}(\mathrm{m}^4 \times \mathrm{n}^{\mathrm{m}})$; DRC and DRC-QM: $\mathrm{O}(\mathrm{n}^{\mathrm{m}})$; Murphy's method: $\mathrm{O}(2^{\mathrm{n}})$; Deng's method and QCI-Fusion: $\max\{\mathrm{O}(\mathrm{n}^{\mathrm{m}}), \mathrm{O}(\mathrm{n}^{\mathrm{n}})\}$. Clearly, PCR5+ and PCR6+ incur a substantial increase in running time when processing large volumes of evidence, rendering them impractical for high real-time requirements. For Wholesale dataset (Table XI), QCI-Fusion achieves the highest average accuracy (72.36%) and F1-score (66.08%), whereas the next-best method reaches only 71.54% accuracy and 64.69% F1-score. For Breast Cancer dataset (Table XII), QCI-Fusion again attains the highest average accuracy (96.9%) and F1-score (96.62%), while the next-best method achieves only 96.8% accuracy and 65.91% F1-score. Meanwhile, increasing λ_{cof} can significantly improve the computational efficiency of QCI-Fusion. Specifically, when $\lambda_{cof}=0.75$the running time of QCI-Fusion is reduced by up to 70% (Wholesale dataset) and by at least 5% (Seeds dataset). Under this setting, QCI tends to classify most QMFs as non-conflicting, thereby minimizing the number of QMFs involved in fusion process. In contrast, when $\lambda_{cof}=0.25$, the accuracy degradation of QCI-Fusion is strictly controlled within 1.3% (Wholesale dataset). In this case, QCI tends to regard most QMFs as conflicting, enabling the majority of QMFs to participate in fusion process. Additionally, for datasets other than Wholesale, QCI-Fusion exhibits identical performance at $\lambda_{cof}=0.75$, $\lambda_{cof}=0.5$ and $\lambda_{cof}=0.25$. This is primarily attributed to the high sensitivity of QCI to discrepancies among QMFs, which causes QCI values to cluster near 0 or 1. Moreover, due to the high consistency among the QMFs generated from these dataset, QCI-Fusion can achieve relatively fast computation at $\lambda_{cof}=0.75$. In summary, QCI-Fusion can be effectively applied to pattern classification tasks, achieving high accuracy and F1-score by eliminating conflicts among QMFs, and its approximation module enables high-throughput real-time deployment.

TABLE XI
THE PERFORMANCE OF RECOGNITION IN SELECTED PERCENTAGES OF WHOLESALE DATASET

Methods	10%		30%		50%		70%		90%		Average		
	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Running time↓
DRC	68.11	65.32	69.63	55.2	69.01	64.74	69.75	59.62	69.43	63.27	69.19	61.63	0.057
Murphy	70.96	68.99	71.42	60.65	71.24	64.59	71.28	57.56	71.63	60.3	71.31	62.42	0.009
PCR5+	70.22	70.65	**73.04**	59.29	66.71	63.51	65.29	56.98	65	59.7	68.05	62.02	71.246
PCR6+	68.72	66.89	67.38	57.25	66.49	63.34	65.29	56.98	69	62.64	67.38	61.42	70.924
DRC-QM	71.3	69.8	71.43	**62.77**	71.6	65.73	71.7	58	71.67	63.24	71.54	63.91	0.057
Deng p=1	71.64	69.88	71.57	59.32	71.74	**66.68**	71.84	**63.49**	71.84	64.08	71.73	64.69	0.117
Deng p=2	71.64	69.96	71.57	58.41	71.74	66.22	71.84	61.29	71.84	64.51	71.73	64.08	0.117
QCI-Fusion $\lambda_{cof} = 0$	**72.38**	**71.74**	71.48	60.57	**73.31**	64.79	**71.93**	62.83	**72.69**	**70.46**	**72.36**	**66.08**	0.118
QCI-Fusion $\lambda_{cof} = 0.25$	72.06	70.08	70.47	60.57	72.61	63.78	70.17	60.83	71.73	69.57	71.41	64.96	0.075 (0.043↓)
QCI-Fusion $\lambda_{cof} = 0.5$	70.60	69.52	69.33	59.98	72.25	62.98	69.07	60.06	71.55	66.72	70.56	63.85	0.056 (0.062↓)
QCI-Fusion $\lambda_{cof} = 0.75$	70.31	68.08	67.29	59.44	71.76	63.22	68.45	58.76	71.17	65.34	69.80	62.97	0.036 (0.082↓)

TABLE XII
THE PERFORMANCE OF RECOGNITION IN SELECTED PERCENTAGES OF BREAST CANCER DATASET

Methods	10%		30%		50%		70%		90%		Average		
	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Acc↑	F1↑	Running time↓
DRC	94.66	89.94	93.75	89.93	93.71	88.66	94.55	90.57	94.5	90.03	94.23	89.83	0.003
Murphy	94.49	94.21	93.95	93.66	93.62	93.29	94.48	94.25	94.48	94.21	94.2	93.92	0.0004
PCR5+	96.92	96.63	**96.66**	**96.36**	96.39	96.05	**97.41**	**97.17**	**96.62**	**96.34**	96.8	96.51	3.128
PCR6+	96.88	96.59	96.62	96.32	96.39	96.05	**97.41**	**97.17**	**96.62**	**96.34**	96.78	96.49	3.014
DRC-QM	95.11	94.83	94.34	94.04	94.17	93.84	94.99	94.76	95.08	94.8	94.74	94.45	0.003
Deng p=1	96.48	96.19	95.54	95.24	95.51	95.17	96.35	96.11	96.3	96.02	96.04	95.75	0.013
Deng p=2	96.53	96.24	95.59	95.29	95.53	95.19	96.02	95.78	96.32	96.04	96	95.71	0.013
QCI-Fusion $\lambda_{cof} = 0$	**97.02**	**96.76**	96.52	96.18	**97.08**	**96.82**	97.25	97	**96.62**	**96.34**	**96.9**	**96.62**	0.013
QCI-Fusion $\lambda_{cof} = 0.25$	97	96.71	96.32	96.02	96.04	95.7	96.88	96.64	96.59	96.31	96.57	96.28	0.011 (0.002↓)
QCI-Fusion $\lambda_{cof} = 0.5$	97	96.71	96.32	96.02	96.04	95.7	96.88	96.64	96.59	96.31	96.57	96.28	0.011 (0.002↓)
QCI-Fusion $\lambda_{cof} = 0.75$	96.49	96.19	95.46	95.15	95.66	95.32	96.4	96.15	96.59	96.31	96.12	95.83	0.008 (0.005↓)

V. FROM PATTERN CLASSIFICATION TO OOD DETECTION

In Section IV-B, QCI-Fusion demonstrates strong performance in classification tasks. However, the traditional classification framework is inherently constrained by the closed-world assumption, which assumes that test samples necessarily belong to the known distribution (in-distribution, ID). This assumption renders QCI-Fusion vulnerable when encountering OOD samples in open-world scenarios. Therefore, this section extends QCI-Fusion from pattern classification to OOD detection, aiming to construct a comprehensive decision-making system that accurately identifies ID samples while robustly rejecting OOD samples.

A. Necessity of Expansion

Penultimate-layer ID features tend to form clusters, while OOD features reside apart [34]. A feasible way to extend QCI-Fusion to OOD detection is to capture feature semantic conflicts induced by heterogeneity. Traditional methods (e.g., Mahalanobis [35], ViM [36] and NCI [34]) treat ID samples as independent entities and compute their feature centers through additive aggregation to form clusters. As a result, local nonlinear interactions for feature channels are smoothed during the summation or projection process. The system may misinterpret severe semantic conflicts among ID features as ordinary statistical fluctuations, ultimately leading to the incorrect decision of OOD samples. In contrast, QDST maps ID features into a common QFoD, represents them as QMFs, and achieves nonlinear semantic aggregation through coherent superposition for decision-making. However, traditional fusion methods, including QCI-Fusion, fail to effectively detect OOD samples. This limitation arises because OOD classes are typically not represented in QFoD, causing these methods to confidently misclassify OOD samples as a specific ID class during decision-making. Although numerous studies in DST have attempted to detect OOD samples [37], [38], methods for OOD detection in QDST remain limited. Therefore, this section presents a novel decision architecture, named QCI-Decision. The detailed procedure is illustrated in Fig. 4. QCI-Decision fuses QMFs within the same ID class via QCI-Fusion to produce feature centers, while the loss of belief flow during fusion and the orthonormalization between test samples and feature centers are explicitly quantified using QCI. QCI-Decision transforms interference from a discrimination burden into a core metric of semantic consistency, thereby providing a reliable foundation for OOD detection.

B. Method Describing

The goal of OOD detection is to develop a score function S that detects OOD samples while preserving the model's original classification capabilities for ID samples. Specifically, given a test sample t, t is classified as OOD iff the score $S(t) > \lambda$, where λ denotes a division threshold.

For a pre-trained network P, this paper considers specific models including VGG [39], WRN [40], ViT [41] and Swin [42]. For detailed settings, please refer to Section V-C1. The QFoD formed by the output channels of P is denoted as $|\Theta\rangle = \{|a_1\rangle, \ldots, |a_n\rangle\}$, where $|a_k\rangle \subseteq |\Theta\rangle$ represents the k-th feature channel. P produces a feature vector l_g for each sample with ground-truth ID label g. Each l_g contains n elements, each corresponding to the activation value of a feature channel $|a_k\rangle \subseteq |\Theta\rangle$.

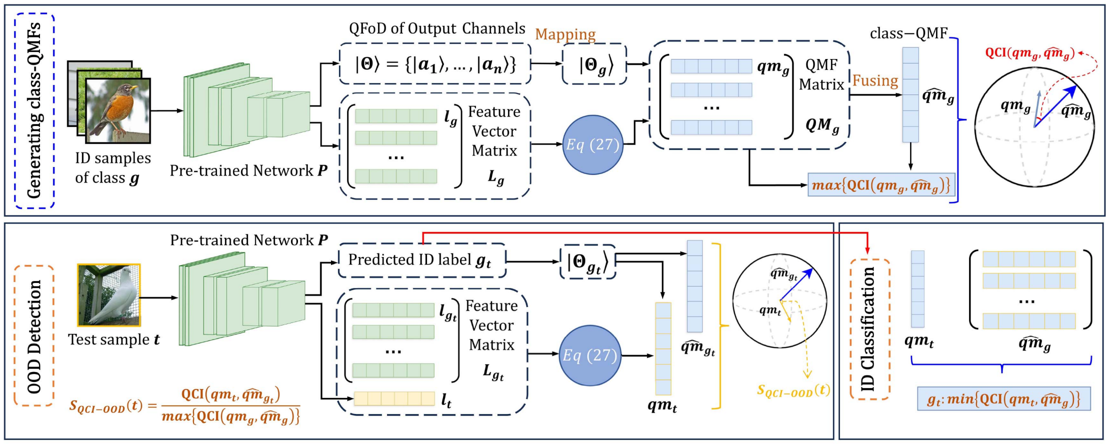


Fig. 4. Overview of The Proposed QCI-Decision.

To mitigate the exponential computational complexity introduced by the power-set operation in QDST [15], QCI-Decision employs a linear channel selection method based on statistical variance for dimension reduction. Specifically, the variance of the column vectors of the matrix L_g, formed by horizontally stacking all l_g, is computed to reflect the fluctuations of $|a_k\rangle$. This method is preferred over alternative dimension reduction techniques for the following reasons: 1) Linear transformations (e.g., PCA [43]) project original features into a principal-component space via coordinate-axis rotation, whereas non-linear transformations (e.g., Isomap [44] and LLE [45]) alter the metric structure of features through manifold mapping. Such transformations introduce semantic interference among the original feature channels, thereby violating the mutual exclusivity principle of elements in QFoD. 2) Samples with the same ID class should exhibit high consistency across feature channels, whereas channels with significant fluctuations are considered individual features of specific samples. These abnormal channels may lead to bias during decision-making and should be disregarded. Therefore, using Eq (24), l_g is mapped into a lower-dimensional QFoD $|\Theta_g\rangle$ to construct QMFs qm_g. Here, $qm_g(|a_k\rangle)$ denotes the quantum basic belief assignment of l_g for the feature channel $|a_k\rangle$.

Subsequently, QCI-Fusion is applied to fuse all qm_g producing the class-QMFs $\widehat{qm}_g$ as feature centers for each ID class. In this process, the conflict detection threshold λ_{cof} functions as a structural filter. Specifically, it performs pre-clustering on highly consistent QMFs, enabling QCI-Fusion to suppress structural noise and enhance the semantic consistency among qm_g. Furthermore, the fusion stage processes qm_g with significant conflicts to obtain the optimal $\widehat{qm}_g$, thereby improving its inclusiveness for ID samples. Intuitively, the feature information of ID samples should closely correspond to class-QMF representations.

For a test sample t, let g_t denote the predicted ID label. Two prediction strategies are considered: 1) The model prediction is used directly, in which case QCI-Decision functions solely as an OOD detector, denoted as QCI-OOD. 2) The prediction is based on class-QMFs. Specifically, the QMF qm_t of t is obtained by mapping the feature vector to $|\Theta_{g_t}\rangle$ using L_g and (24). The ID label is then determined by selecting the class that yields the minimum conflict, i.e., $\min\{\mathrm{QCI}(qm_t, \widehat{qm}_g)\}$. For OOD detection, the scoring function is defined as:

$$S_{QCI-OOD}(t) = \frac{\mathrm{QCI}(qm_t, \widehat{qm}_{g_t})}{\max\{\mathrm{QCI}(qm_g, \widehat{qm}_{g_t})\}}, \tag{28}$$

where $\max\{\mathrm{QCI}(qm_g, \widehat{qm}_{g_t})\}$ denotes the belief flow loss for class g, and $\widehat{qm}_{g_t}$ denotes the class-QMF associated with the label g_t.

C. Experiments

In this section, QCI-Decision is evaluated on both ID classification and OOD detection across multiple benchmark datasets.

1) Experiment Settings: Datasets: To conduct a comprehensive benchmark evaluation of QCI-OOD, we use Mini-ImageNet and CIFAR-10 as the ID datasets. In QCI-OOD, class-QMFs are constructed from the training and validation images and evaluated on the test images. Mini-ImageNet is a subset of ImageNet-1k [46] comprising 100 classes, each containing 384 training images, 96 validation images, and 120 test images. We employ five OOD datasets whose labels do not overlap with those of the ID dataset: 1) iNaturalist is a fine-grained species classification dataset; here, we use the subset from [47], which contains 10,000 test images. 2) Texture [48] consists of 5,640 natural texture images. 3) ImageNet-O [49] is specifically designed to evaluate the robustness of visual models and contains 2,000 test images. 4) SUN [47] is a scene image dataset, and we use a subset containing 10,000 test images. 5) Places [47] is a dataset based on human visual cognition principles, and we use a subset containing 10,000 test images. CIFAR-10 [50] is a widely used benchmark comprising 10 classes, each containing 6,000 training images and 1,000 test images. We randomly split the original training set into 5,000 images for training and 1,000 for validation. We employ five OOD datasets: 1) CIFAR-100 [50] contains 100 classes that are semantically distinct from the CIFAR-10 classes. 2) Texture, as described above. 3) Tiny-ImageNet [51] is a downsampled version of ImageNet-1k containing 10,000 test images. 4) Svhn [52] consists of real-world

TABLE XIII
PERFORMANCES OF QCI-DECISION AND THE BASELINE METHODS IN OOD DETECTION, WHERE ID DATASET IS MINI-IMAGENET. SYMBOL $*$ INDICATES THAT ID SAMPLES ARE REQUIRED.

Model	Method	iNaturalist		Texture		ImageNet-O		Sun		Places		Average		Acc↑
		AUC↑	FPR↓	AUC↑	FPR↓	AUC↑	FPR↓	AUC↑	FPR↓	AUC↑	FPR↓	AUC↑	FPR↓	
Diffusion (need train)	DiffPath-6D	58.72	97.71	74.55	72.83	69.79	78.55	53.5	96.56	53.33	97.03	61.98	88.54	–
	EigenScore	54.48	96.19	77.97	69.95	65.98	83.57	58.79	92.13	58.21	92.78	63.09	86.92	–
VGG	MSP	89.63	40.68	86.12	51.47	56.01	84.01	81.63	59.17	81.04	58.13	78.89	58.69	89.76
	Energy	92.49	28.11	92.05	34.73	64.55	75.66	87.75	44.99	84.99	51.64	84.37	47.03	89.76
	ODIN	89.13	37.75	88.52	40.38	82.52	56.38	85.63	43.71	84.99	45.55	86.16	44.75	89.76
	Mahalanobis*	91.74	29.63	91.83	32.43	84.59	51.11	89.15	**35.89**	**90.14**	**35.88**	89.49	36.99	89.76
	ViM*	92.28	23.32	93.1	**22.81**	91.72	33.83	88.52	38.15	89.12	41.43	90.98	31.91	89.76
	NCI*	**95.48**	**17.64**	**94.08**	23.99	87.69	38.65	90.08	36.16	88.27	38.84	91.12	31.24	89.76
	QCI-OOD*	95.43	18.63	92.9	23.27	**92.38**	**32.96**	**90.31**	38.43	89.91	39.26	**92.19**	**30.51**	89.76
	QCI-Decision*	94.52	19.55	92.46	29.97	91.76	34.15	89.54	37.86	89.29	39.35	91.51	32.18	88.27
WRN	MSP	95.01	24.92	89.25	48.7	63.83	84.25	89.62	45.84	88.99	46.24	85.34	49.99	95.84
	Energy	95.3	18.91	90.7	45.94	71.01	81.17	91.04	39.65	89.71	45.2	87.55	46.17	95.84
	ODIN	95.5	18.63	91.99	45.96	74.02	79.57	90.8	39.08	89.32	46.11	88.32	45.87	95.84
	Mahalanobis*	96.79	17.31	97.02	12.53	93.91	25.51	92.9	30.56	92.03	34.04	94.53	23.99	95.84
	ViM*	96.94	13.56	98.39	7.26	95.61	25.99	94.59	21.29	94.13	22.03	95.93	18.03	95.84
	NCI*	98.31	**4.73**	98.5	6.39	95.79	24.18	**96.6**	**14.74**	95.24	19.89	**96.89**	14	95.84
	QCI-OOD*	**98.49**	5.1	98.29	5.8	**95.88**	**19.15**	95.83	16.41	95.46	18.53	96.8	**13**	95.84
	QCI-Decision*	97.87	8.49	**98.59**	**5.06**	94.95	19.87	95.23	15.55	**95.57**	**17.55**	96.44	13.3	95.41
ViT	MSP	91.42	44.95	83.92	58.56	73.15	83.5	84.93	56.63	83.56	58.19	83.4	60.37	93.25
	Energy	89.16	56.36	84.12	57.23	72.11	89.65	84.16	59.95	84.44	56.89	82.8	64.02	93.25
	ODIN	89.74	50.58	90.16	43.11	84.15	70.15	87.4	46.85	87.15	50.18	87.72	52.17	93.25
	Mahalanobis*	96.28	10.23	92.85	23.38	93.26	25.3	91.95	37.46	92.41	38.69	93.35	27.01	93.25
	ViM*	97.99	7.38	96.01	14.33	94.86	22.91	93.16	35.45	93.05	35.91	95.01	23.2	93.25
	NCI*	98.47	6.57	**96.9**	14.08	**95.84**	**16.58**	95.53	19.2	95.03	21.15	96.35	15.52	93.25
	QCI-OOD*	**98.69**	**4.19**	96.79	10.48	95.5	17.23	**96.28**	**17.85**	**95.64**	**20.49**	**96.58**	**14.04**	93.25
	QCI-Decision*	98.3	4.31	96.51	**10.05**	95.27	16.97	95.53	18.86	94.98	22.45	96.12	14.53	92.54
Swin	MSP	95.67	18.73	88.66	44.76	76.15	74.88	89.02	49.31	88.78	49.01	87.65	47.34	96.75
	Energy	96.51	15.45	92	37.23	80.77	68.01	91.52	43.42	87.11	51.14	89.58	43.05	96.75
	ODIN	96.48	15.79	92.92	35.41	87.38	54.27	90.07	47.04	88.93	48.63	91.16	40.23	96.75
	Mahalanobis*	**99.53**	**2.19**	96.21	11.88	95.07	29.98	96.85	14.83	96.19	17.99	96.77	15.37	96.75
	ViM*	98.46	6.1	97.8	8.53	95.45	26.41	97.16	14.44	96.68	16.31	97.11	14.36	96.75
	NCI*	98.45	6.92	97.8	8.64	**96.77**	**13.16**	97.8	**10.63**	96.21	15.44	97.4	10.96	96.75
	QCI-OOD*	98.98	3.64	97.67	6.46	96.42	13.87	96.73	13.82	**97.49**	**10.26**	**97.46**	**9.61**	96.75
	QCI-Decision*	98.89	3.87	**97.85**	**5.76**	95.99	13.66	96.43	14.09	96.82	14.62	97.2	10.4	96.59

house number images collected from Google Street View and contains 26,032 test images. 5) Places365 [53] is a large-scale scene recognition dataset containing 36,500 test images.

Compared methods: All experiments are implemented using Python 3.10 and PyTorch 2.1.0. Experiments are conducted on a single NVIDIA GeForce RTX 4090 GPU with CUDA version 12.1. Our method is compared with six post-hoc baselines that do not require fine-tuning, including MSP [54], Energy [55], ODIN [56], Mahalanobis [35], ViM [36], and NCI [34]. All these methods are evaluated using the OpenOOD library [51] with their default configurations. Additionally, we include two generative OOD detection methods, namely DiffPath-6D [57] and EigenScore [58], for comparison. On CIFAR-10, we directly evaluate using the released pretrained weights. On Mini-ImageNet, we fine-tune the publicly available checkpoints using the default training configuration.

Evaluation metrics: Acc (%) is used to evaluate the classification performance on ID samples. For OOD detection, two commonly used metrics are adopted [59]: 1) AUC (%) is a threshold-free metric that computes the area under the receiver operating characteristic curve. Higher values indicate better performance. 2) FPR (%) denotes FPR@TPR95, i.e., the false positive rate when the true positive rate reaches 95%. Lower values indicate better performance.

Experiment settings: We conduct experiments on two CNN-based models: VGG [39] and WRN [40], two Transformer-based models: ViT [41] and Swin [42]. For Mini-ImageNet, the pre-trained VGG16, WideResNet50-2, ViT-B-16 and Swin-B models [2] from ImageNet-1 K are used, and only the decision layers are fine-tuned on Mini-ImageNet using an SGD optimizer with a learning rate of 0.001, a cross-entropy loss, a batch size of 64, a momentum of 0.9, and a weight decay of 5×10^{-4}. During FP32 training, random horizontal flipping is applied for data augmentation and the weights from the 100-th epoch are selected. For CIFAR-10, models are trained from scratch using an SGD optimizer with an initial learning rate of 0.01, a momentum of 0.9, and a weight decay of 5×10^{-4}. All input images are upsampled to 224×224. The model is trained for 200 epochs with a batch size of 64, and the learning rate is decayed following a cosine annealing schedule. During training, standard data augmentation including random cropping with padding and random horizontal flipping is applied. We report results using the model weights from the final epoch. The detection threshold λ_{cof} is set to 0.5. Meanwhile, the number of elements in QFoD $|\Theta_g\rangle$ is set to VGG:2560, WRN:1024, ViT:512 and Swin:512, respectively.

2) Results of Comparative Experiments: We report the results on Mini-ImageNet (Table XIII) and CIFAR-10 (Table XIV). The best AUC and FPR values are highlighted in bold, and the second-best results are marked with underlines.

ID classification ability of QCI-Decision: The results show that QCI-Decision introduces only negligible changes to the original classification accuracy of each model. On

[2]https://pytorch.org/vision/stable/models.html

TABLE XIV
PERFORMANCES OF QCI-DECISION AND THE BASELINE METHODS IN OOD DETECTION, WHERE ID DATASET IS CIFAR-10. SYMBOL ∗ INDICATES THAT ID SAMPLES ARE REQUIRED.

Model	Method	CIFAR-100		Texture		Tiny-ImageNet		Svhn		Places365		Average		Acc↑
		AUC↑	FPR↓	AUC↑	FPR↓	AUC↑	FPR↓	AUC↑	FPR↓	AUC↑	FPR↓	AUC↑	FPR↓	
Diffusion (need train)	DiffPath-6D	58.98	89.15	92.4	30.79	56.38	90.8	96.83	17.18	75.79	76.71	76.08	60.93	–
	EigenScore	82.55	61.2	78.5	80.91	73.97	69.93	73.92	72.58	76.54	76.73	77.1	72.27	–
VGG	MSP	84.25	52.86	84.18	49.3	81.24	59.34	87.34	34.32	83.52	55.93	84.11	50.35	90.12
	Energy	**85.44**	54.75	84.91	47.45	81	58.91	89.77	35.48	84.68	53.62	85.16	50.04	90.12
	ODIN	84.29	54.07	85.82	**46.24**	83.09	55.64	88.54	32.95	84.8	52	85.31	48.18	90.12
	Mahalanbois∗	82.67	57.05	**86.03**	57.76	81.6	57.2	87.48	39.66	80.57	57.58	83.67	53.85	90.12
	ViM∗	81.85	55.43	83.05	53.89	82.17	56.17	88.77	34.55	84.89	51.44	84.15	50.3	90.12
	NCI∗	83.58	56.12	83.23	50.64	83.46	57.12	**93.41**	**26.94**	82.89	66.81	85.31	51.53	90.12
	QCI-OOD∗	85.36	**50.62**	82.38	53.15	**84.62**	**48.02**	89.8	32.64	**86.98**	**48.33**	**85.83**	**46.55**	90.12
	QCI-Decision∗	85.32	51.53	82.21	52.97	84.49	49.4	87.89	36.63	86.92	47.35	85.37	47.58	89.2
WRN	MSP	84.96	46.43	89.82	33.89	84.49	49.86	89.4	24.05	84.58	50.03	86.65	40.85	93.85
	Energy	85.93	**44.46**	90.2	32.83	85.33	48.24	90.06	**21.3**	86.1	49.42	87.52	39.25	93.85
	ODIN	84.95	46.99	90.55	33.98	84.05	50.84	89.72	24.18	84.1	50.39	86.67	41.28	93.85
	Mahalanobis∗	83.44	48.97	90.04	37.5	84.62	54.92	89.91	29.91	87.8	44.17	87.16	43.09	93.85
	ViM∗	85.77	47.85	**92.35**	28.69	84.39	50.62	**91.83**	22.63	83.67	51.34	87.6	40.23	93.85
	NCI∗	**86.02**	46.92	87.86	32.36	86.48	**47.14**	90.12	23.71	**89.49**	**39.5**	87.99	**37.93**	93.85
	QCI-OOD∗	85.7	45.41	91.32	**28.15**	86.53	49.76	90.27	24.67	89.13	41.95	**88.59**	37.99	93.85
	QCI-Decision∗	85.67	44.54	90.27	28.34	**86.59**	50.52	90.25	23.67	89.09	41.59	88.37	37.73	93.57
ViT	MSP	87.17	41.6	91.92	23.53	86.23	47.1	94.76	22.08	87.1	43.87	89.44	35.64	92.8
	Energy	87.39	39.49	92.02	22.88	87.87	44.88	95.65	21.95	88.58	42.38	90.3	34.32	92.8
	ODIN	87.44	44.88	94.39	24.67	86.1	51.64	93.05	34.3	87.06	48.62	89.61	40.82	92.8
	Mahalanbois∗	89.86	42.26	95.35	22.19	83.89	55.2	98.36	8.92	88.13	46.43	91.12	35	92.8
	ViM∗	91.61	**36.31**	**98.65**	**7.35**	**91.32**	**38.89**	98.4	7.53	90.55	43.15	**94.11**	26.65	92.8
	NCI∗	90.01	43.22	97.2	13.62	88.95	49.76	98.36	6.32	90.11	46.65	92.93	31.91	92.8
	QCI-OOD∗	**92.19**	36.52	97.66	8.17	90.24	41.76	**98.48**	**4.64**	**91.68**	**38.97**	94.05	**26.01**	92.8
	QCI-Decision∗	92.11	36.39	97.63	8.07	90.28	40.78	98.35	5.72	91.43	39.47	93.96	26.09	92.48
Swin	MSP	89.15	38.59	92.99	23.35	88.11	44.3	94.95	19.05	89.07	38.63	90.85	32.78	95.11
	Energy	89.7	39.08	92.1	23.47	89.19	46.78	96.29	17.14	90.38	34.74	91.53	32.24	95.11
	ODIN	89.41	40.37	94.72	23.53	87.93	49.12	93.88	27.96	89.26	42.72	91.04	36.74	95.11
	Mahalanbois∗	87.52	53.69	98.12	10.48	87.16	52.9	98.82	5.89	89.05	43.09	92.13	33.21	95.11
	ViM∗	92.57	34.26	**98.97**	**5.24**	**92.22**	35.82	**99.33**	**2.96**	91.33	28.17	94.88	21.29	95.11
	NCI∗	91.65	37.96	97.64	12.08	90.74	43.79	98.23	8.69	91.67	26.42	93.99	25.79	95.11
	QCI-OOD∗	**93.12**	**33.34**	98.51	9.72	91.68	**31.67**	98.9	3.45	**93.08**	**25.06**	**95.04**	**20.65**	95.11
	QCI-Decision∗	93.06	33.63	98.59	9.8	91.6	31.7	98.81	3.53	92.05	30.16	94.82	21.76	94.91

Mini-ImageNet, the observed performance decreases are VGG:-1.49%, WRN:-0.43%, ViT:-0.71% and Swin:-0.16%. On CIFAR-10, the accuracy decreases are even smaller (VGG:-0.92%, WRN:-0.28%, ViT:-0.32% and Swin:-0.20%). QCI-Decision differs from QCI-Fusion, which models QFoD using labels, by performing decision-making based on feature representations. This eliminates the strong QFoD constraints introduced by direct label-space mapping, thereby enabling flexible adaptation to different label spaces without redefining QFoD. Overall, the results demonstrate the strong robustness of QCI-Decision in ID classification.

OOD detection ability of QCI-Decision: QCI-OOD, the OOD detector of QCI-Decision, achieves the best or second-best AUC and FPR in almost all settings compared with post-hoc OOD detection methods. For Mini-ImageNet, on the VGG backbone, QCI-OOD outperforms the second-best method (NCI) with the largest average AUC improvement of 1.07%. It also brings consistent gains on ViT (+0.23%) and Swin (+0.06%) compared with the second-best methods. The only exception occurs on WRN, where QCI-OOD is slightly inferior to NCI with a marginal AUC drop of 0.09%. In terms of FPR, although QCI-OOD is higher than NCI by 1% on WRN, it achieves the lowest average FPR on the other three backbones, reducing FPR by 1.03% (VGG), 1.48% (ViT), and 1.35% (Swin) compared with the second-best methods. For CIFAR-10, QCI-OOD again demonstrates strong competitiveness. On WRN, it achieves the largest AUC improvement over NCI (+0.6%). It also improves over the second-best methods on VGG (+0.22%) and Swin (+0.16%). The only slight degradation appears on ViT, where QCI-OOD is lower than ViM by 0.06% in AUC. Regarding FPR, although QCI-OOD is higher than ViM by 0.64% on ViT, it achieves the lowest FPR on VGG and Swin, reducing FPR by 1.63% and 0.64%, respectively. Notably, two generative OOD detection methods, DiffPath-6D and EigenScore, both achieve lower performance than the VGG-based QCI-OOD on both Mini-ImageNet and CIFAR-10 datasets. Compared with directly using model predictions, QCI-Decision—when predicting ID labels using class-QMFs—still maintains a strong capability for detecting OOD samples. This result further demonstrates the robustness of the proposed QCI-Decision in OOD detection.

D. Mechanistic and Parameter Analysis

This section provides a detailed analysis of the mechanisms and parameter selection of QCI-Decision.

1) Impact of Feature Channel Activation of Models: To analyze the compatibility between different models and QCI-Decision, we examine the activation patterns of channels in the penultimate layer. The following metrics are considered: 1) Activation ratio is the proportion of elements exceeding a threshold, used to measure representation density and potential redundancy. 2) Average amplitude and standard deviation characterize activation intensity and inter-channel dispersion. 3) Number of activation channels is the average number of

TABLE XV
STATISTICAL METRICS OF PENULTIMATE-LAYER ACTIVATION CHANNELS IN DIFFERENT BACKBONES

Backbones (dimension)	Activation ratio	Average amplitude	Average standard	Number of active channels	Channel variance	Average QCI of class-QMFs
VGG (4096)	0.999977	0.2881	0.2282	4095.91	0.0483	0.4776
WRN (2048)	0.999962	0.2032	0.2058	2047.92	0.041	0.8993
ViT (768)	0.999989	0.5763	0.4657	767.99	0.2051	0.9656
Swim (1024)	0.999951	0.1276	0.0985	1023.94	0.0093	0.9041

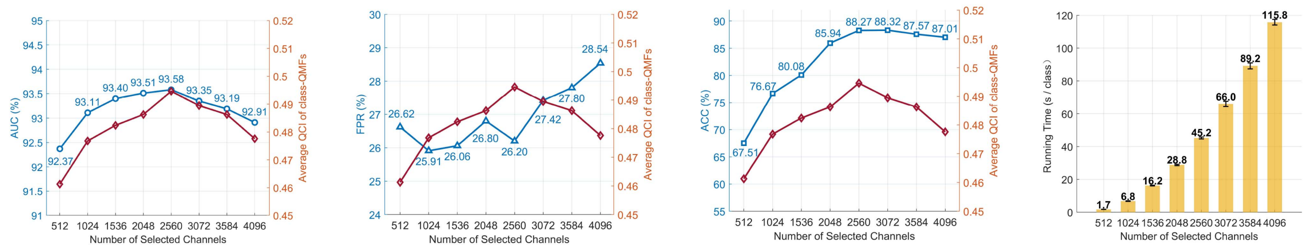


Fig. 5. Abortion of Liner Channel Selection in default settings.

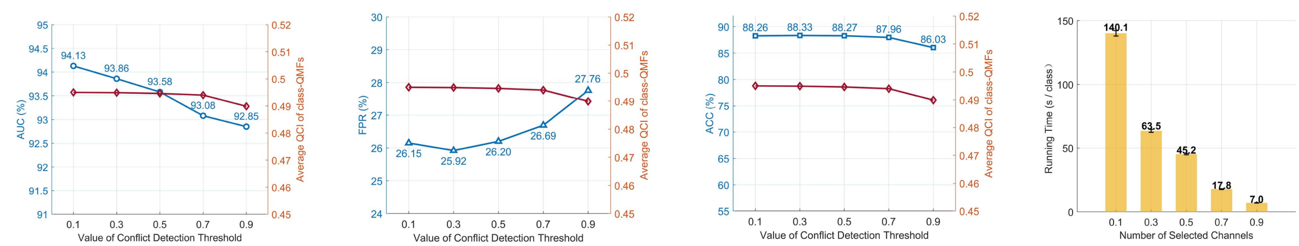


Fig. 6. Abortion of Conflict Detection Threshold in default settings.

activation channels per sample, reflecting the effective activation channel dimension. 4) Channel variance measures channel selectivity. Larger variance indicates lower redundancy and stronger discriminability.

Table XV shows that all backbones exhibit an extremely high activation ratio (> 0.9999), when the activation threshold is set to 10^{-5}. This indicates that deep features are globally dense and redundant. Meanwhile, ViT exhibits the highest amplitude (0.5763) and channel variance (0.2051), suggesting stronger feature heterogeneity. For QMFs, the statistical metrics of activation channels directly determine the ability to construct discriminative quantum states in the Hilbert space. The performance of QCI-Decision depends on the spatial expansion capability of backbones in the Hilbert space. When a high-dimensional backbone (e.g., VGG) exhibits low inter-channel variance, the information contributed by each channel lacks sufficient distinctiveness and phase diversity. This is because excessive dimension disperses the belief flow to most channels, weakening the orthonormalization among class-QMFs. Thus, its measured QCI (0.4776) becomes significantly smaller than that of the other three backbones. In contrast, models with moderate dimension and high heterogeneity (e.g., ViT) show better compatibility with QCI-Decision.

2) Impact of Liner Channel Selection: Based on the redundancy analysis in Section V-D1, Fig. 5 investigates the impact of channel selection on the performance of QCI-Decision. 1) ID classification performance (ACC): When the number of selected channels decreases from 4096 to 2560, ACC increases from 87.01% to 88.27%, indicating that QCI-Decision effectively filters channel noise. However, when the number of channels is further reduced to 512, ACC drops sharply to 65.51%, as excessive reduction leads to the loss of essential semantic information 2) OOD detection performance (AUC and FPR): Here, model prediction is employed. The AUC increases from 92.91% at 4096 to a peak of 93.58% at 2560, where the QCI value reaches its maximum, indicating the highest relative orthogonality among class-QMFs. From 1024 to 2560, QCI-Decision maintains stable performance (AUC $> 93.3\%$), demonstrating strong robustness during hyperparameter tuning. 3) Running time: The reported running time corresponds to the fusion process of a single class (96 QMFs). At the optimal point (2560), the running time is 45.22 s, achieving a 3$\times$ speedup compared with the 4096 version (115.79 s). When selected channels further reduce to 512, the running time decreases to 16.23 s. For the fusion process in QCI-Decision, i.e., QCI-Fusion, the complexity is $\max\{\mathrm{O}(\mathrm{n}^{\mathrm{m}}), \mathrm{O}(\mathrm{n}^{\mathrm{n}})\}$. Channel selection effectively reduces the feature dimension (i.e., n) and serves as a form of model compression that lowers computational overhead.

3) Impact of Conflict Detection Threshold: Since all models exhibit similarly dense activation patterns, we analyze the effect of λ_{cof} using VGG in Fig. 6, and the conclusions can be generalized to other models. When λ_{cof} increases from 0.1 to 0.9, the ID classification accuracy decreases from 88.26% to 86.03%. Meanwhile, the OOD detection performance also degrades, with the AUC decreasing from 94.13% to 92.85% and the FPR increasing from 26.15% to 27.76%. This behavior indicates that a larger threshold strengthens the aggregation of "low-conflict" QMFs, which reduces the diversity of ID feature information and suppresses the heterogeneity that is important for distinguishing ID and OOD samples. In contrast, the running time decreases significantly as λ_{cof} increases. The running time drops from 140.08s at $\lambda_{cof} = 0.1$ to 7.03s at $\lambda_{cof} = 0.9$, corresponding to approximately a $20\times$ speedup. Since the number of QMFs involved in fusion is a primary factor affecting the computational complexity of QCI-Decision, increasing λ_{cof} can effectively limit it. Finally, we set $\lambda_{cof} = 0.5$ because QCI-Decision naturally constrains most QCI values near the two extremes of the curve (close to 0 or 1), representing highly consistent or strongly conflicting QMFs. This threshold provides a natural separation boundary that preserves meaningful structural differences while maintaining efficient fusion.

VI. Conclusion

This paper proposes a quantum conflict indicator (QCI) to effectively manage conflicts among multiple QMFs. QCI is the first metric to satisfy ideal conflict measurement properties. Building on this, a novel quantum conflict fusion method (QCI-Fusion) is introduced. Across various commonly used pattern classification tasks, QCI-Fusion achieves the highest classification accuracy compared to other fusion methods. Furthermore, this study proposes QCI-Decision, an extension of the QCI-Fusion framework, which models QMFs using feature channels and performs decision-making based on the conflict between QMFs. Experimental results show that the classification accuracy of QCI-Decision deviates from the original model prediction by at most 1.49%, with smaller deviations observed for stronger models. Moreover, compared with the best baseline anomaly detection method, QCI-Decision improves AUC by up to 0.6% and reduces FPR by up to 1.63%.

Meanwhile, the optimization of approximate computation and linear channel selection improves the deployability of QCI-Decision in practical scenarios such as autonomous systems and medical diagnostics. Heterogeneous data (e.g., LiDAR/Camera or CT/MRI) can be mapped into QMFs through evidence encoders or generative algorithms. Under limited labeled data, QCI-Fusion can be directly applied for decision-making, whereas QCI-Decision is suitable for open environments with unknown categories (e.g., unseen road conditions or rare diseases). Future work will also explore quantum circuit acceleration [60] to further improve computational efficiency and real-time scalability.

References

[1] B. Kosko, "Counting with fuzzy sets," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. TPAMI-8, no. 4, pp. 556–557, Jul. 1986.

[2] K. Daijin and B. Y. Sung, "A handwritten numeral character classification using tolerant rough set," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 22, no. 9, pp. 923–937, Sep. 2000.

[3] L. A. Zadeh, "A note on z-numbers," *Inf. Sci.*, vol. 181, no. 14, pp. 2923–2932, 2011.

[4] G. Shafer, "Dempster-shafer theory," in *Encyclopedia of Artificial Intelligence*, vol. 1. London, U.K.: Bloomsbury Publishing, 1992, pp. 330–331.

[5] X. Deng and W. Jiang, "A total uncertainty measure for d numbers based on belief intervals," *Int. J. Intell. Syst.*, vol. 34, no. 12, pp. 3302–3316, 2019.

[6] R. R. Yager, "Entropy and specificity in a mathematical theory of evidence," *Int. J. Gen. Syst.*, vol. 9, no. 4, pp. 249–260, 1983.

[7] Z. Gao et al., "Environmental assessment under uncertainty using Dempster–Shafer theory and z-numbers," *J. Ambient Intell. Humanized Comput.*, vol. 11, pp. 2041–2060, 2020.

[8] J. Geng, W. Jiang, K. Huang, and X. Deng, "Multi-scale metric learning for few-shot learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 3, pp. 1091–1102, Mar. 2021.

[9] G. Zhao, Q. Feng, C. Chen, Z. Zhou, and Y. Yu, "Diagnose like a radiologist: Hybrid neuro-probabilistic reasoning for attribute-based medical image diagnosis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7400–7416, Nov. 2022.

[10] F. Xiao, "GIQ: A generalized intelligent quality-based approach for fusing multisource information," *IEEE Trans. Fuzzy Syst.*, vol. 29, no. 7, pp. 2018–2031, Jul. 2021.

[11] E. Herrera-Viedma et al., "Revisiting fuzzy and linguistic decision making: Scenarios and challenges for making wiser decisions in a better way," *IEEE Trans. Syst., Man, Cybern.: Syst.*, vol. 51, no. 1, pp. 191–208, Jan. 2021.

[12] L. Pan and Y. Deng, "A new complex evidence theory," *Inf. Sci.*, vol. 608, pp. 251–261, 2022.

[13] L. Xiong, X. Su, and H. Qian, "Conflicting evidence combination from the perspective of networks," *Inf. Sci.*, vol. 580, pp. 408–418, 2021.

[14] Z.-G. Liu, G.-H. Qiu, S.-Y. Wang, T.-C. Li, and Q. Pan, "A new belief-based bidirectional transfer classification method," *IEEE Trans. Cybern.*, vol. 52, no. 8, pp. 8101–8113, Aug. 2022.

[15] X. Gao, L. Pan, and Y. Deng, "A distance of quantum mass function and its application in multi-source information fusion method based on discount coefficient," *Eng. Appl. Artif. Intell.*, vol. 116, pp. 1–11, 2022.

[16] W. Fan and F. Xiao, "A complex Jensen–Shannon divergence in complex evidence theory with its application in multi-source information fusion," *Eng. Appl. Artif. Intell.*, vol. 116, pp. 1–15, 2022.

[17] J. Huang and F. Xiao, "Complex belief divergence measures for multi-source information fusion," *IEEE Trans. Syst., Man, Cybern.: Syst.*, vol. 55, no. 2, pp. 948–962, Feb. 2025.

[18] L. Gao, F. Xiao, and D. Pelusi, "A complex belief $\mathcal{X}^2$ divergence in complex evidence theory and its application for pattern classification," *Eng. Appl. Artif. Intell.*, vol. 126, pp. 1–15, 2023.

[19] Y. Deng, "Uncertainty measure in evidence theory," *Sci. China Inf. Sci.*, vol. 63, no. 11, 2020, Art. no. 210201.

[20] F. Xiao, Z. Cao, and A. Jolfaei, "A novel conflict measurement in decision-making and its application in fault diagnosis," *IEEE Trans. Fuzzy Syst.*, vol. 29, no. 1, pp. 186–197, Jan. 2021.

[21] W. Liu, "Analyzing the degree of conflict among belief functions," *Artif. Intell.*, vol. 170, no. 11, pp. 909–924, 2006.

[22] S. Destercke and T. Burger, "Toward an axiomatic definition of conflict between belief functions," *IEEE Trans. Cybern.*, vol. 43, no. 2, pp. 585–596, Apr. 2013.

[23] F. Xiao, "CED: A distance for complex mass functions," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 4, pp. 1525–1535, Apr. 2021.

[24] F. Xiao, "CEQD: A complex mass function to predict interference effects," *IEEE Trans. Cybern.*, vol. 52, no. 8, pp. 7402–7414, Aug. 2022.

[25] W. Liu, "Analyzing the degree of conflict among belief functions," *Artif. Intell.*, vol. 170, no. 11, pp. 909–924, 2006.

[26] Q. Wang and Z. Shen, "UBETI: A dv-hop localization algorithm for IoT with distance upper bound estimation by triangle inequality," *IEEE Internet Things J.*, vol. 12, no. 3, pp. 2445–2457, Feb. 2025.

[27] W. Jiang, "A correlation coefficient for belief functions," *Int. J. Approx. Reasoning*, vol. 103, pp. 94–106, 2018.

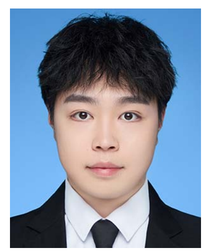
[28] F. Xiao and W. Pedrycz, "Negation of the quantum mass function for multisource quantum information fusion with its application to pattern classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2054–2070, Feb. 2023.

[29] H. Garg and D. Rani, "Multi-criteria decision making method based on Bonferroni mean aggregation operators of complex intuitionistic fuzzy numbers," *J. Ind. Manage. Optim.*, vol. 17, no. 5, pp. 2279–2306, 2021.

[30] F. Xiao, Z. Cao, and T. C. Lin, "A complex weighted discounting multisource information fusion with its application in pattern classification," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 8, pp. 7609–7623, Aug. 2023.

[31] G. Costantini, D. Casali, and M. Carota, "A pattern classification method based on a space-variant CNN template," in *Proc. 10th Int. Workshop Cellular Neural Netw. Appl.*, 2006, pp. 1–5.
[32] C. K. Murphy, "Combining belief functions when evidence conflicts," *Decis. Support Syst.*, vol. 29, no. 1, pp. 1–9, 2000.
[33] T. Dezert, J. Dezert, and F. Smarandache, "Improvement of proportional conflict redistribution rules of combination of basic belief assignments," in *Proc. Adv. Appl. DSmT Inf. Fusion*, 2023, vol. 85, pp. 1–27.
[34] L. Liu and Y. Qin, "Detecting out-of-distribution through the lens of neural collapse," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2025, pp. 15424–15433.
[35] K. Lee, H. Lee, K. Lee, and J. Shin, "A simple unified framework for detecting out-of-distribution samples and adversarial attacks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, vol. 31, pp. 1–11.
[36] H. Wang, L. Feng, Z. Li, and W. Zhang, "ViM: Out-of-distribution with virtual-logit matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 4921–4930.
[37] P. Smets and R. Kennes, "The transferable belief model," *Artif. Intell.*, vol. 66, no. 2, pp. 191–234, 2000.
[38] W. Bao, Q. Yu, and Y. Kong, "Evidential deep learning for open set action recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 13349–13358.
[39] J. Kim, K. J. Lee, and K. M. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 1646–1654.
[40] S. Zagoruyko and N. Komodakis, "Wide residual networks," in *Proc. Brit. Mach. Vis. Conf.*, 2016, pp. 1–15.
[41] X. Wang, Y. Fang, R. Wu, and W. Liu, "What makes for hierarchical vision transformer?," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12714–12720, Oct. 2023.
[42] Y. Rao, Z. Liu, W. Zhao, J. Zhou, and J. Lu, "Dynamic spatial sparsification for efficient vision transformers and convolutional neural networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10883–10897, Sep. 2023.
[43] H. Abdi and L. J. Williams, "Principal component analysis," *Wiley Interdiscipl. Rev.: Comput. Statist.*, vol. 2, no. 4, pp. 433–459, 2010.
[44] J. B. Tenenbaum, V. Silva, and J. C. Langford, "A global geometric framework for nonlinear dimensionality reduction," *Science*, vol. 290, no. 5500, pp. 2319–2323, 2000.
[45] S. T. Roweis and L. K. Saul, "Nonlinear dimensionality reduction by locally linear embedding," *Science*, vol. 290, no. 5500, pp. 2323–2326, 2000.
[46] J. Deng et al., "Imagenet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2009, pp. 248–255.
[47] R. Huang and Y. Li, "MOS: Towards scaling out-of-distribution detection for large semantic space," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 8710–8719.
[48] M. Cimpoi et al., "Describing textures in the wild," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2014, pp. 3606–3613.
[49] D. Hendrycks et al., "Natural adversarial examples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 15262–15271.
[50] A. Krizhevsky and G. Hinton, "Learning multiple layers of features from tiny images," 2009. [Online]. Available: https://api.semanticscholar.org/CopusID:18268744
[51] J. Yang et al., "OpenOOD:Benchmarking generalized out-of-distribution detection," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 32598–32611.
[52] Y. Netzer et al., "Reading digits in natural images with unsupervised feature learning," in *Proc. NIPS Workshop Deep Learn. Unsupervised Feature Learn.*, 2011, pp. 1–9.
[53] B. Zhou, A. Lapedriza, A. Khosla, A. Oliva, and A. Torralba, "Places: A 10 million image database for scene recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 6, pp. 1452–1464, Jun. 2018.
[54] H. Dan and G. Kevin, "A baseline for detecting misclassified and out-of-distribution examples in neural networks," in *Proc. Int. Conf. Learn. Representations*, 2018, pp. 1–12.
[55] J. Owens, W. Liu, X. Wang, and Y. Li, "Energy-based out-of-distribution detection," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, vol. 33, pp. 21464–21475.
[56] S. Liang, Y. Li, and R. Srikant, "Enhancing the reliability of out-of-distribution image detection in neural networks," in *Proc. Int. Conf. Learn. Representations*, 2018, pp. 1–15.
[57] A. Heng and H. Soh, "Out-of-distribution detection with a single unconditional diffusion model," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, vol. 37, pp. 43952–43974.
[58] S. Shoushtari et al., "EigenScore: OOD detection using posterior covariance in diffusion models," in *Proc. Int. Conf. Learn. Representations*, 2025, pp. 1–23.
[59] K. Zhou, Z. Liu, Y. Qiao, T. Xiang, and C. C. Loy, "Domain generalization: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4396–4415, Apr. 2023.
[60] Q. Zhou, G. Tian, and Y. Deng, "BF-QC: Belief functions on quantum circuits," *Expert Syst. Appl.*, vol. 223, 2023, Art. no. 119885.

Yilin Dong was born in Yancheng, China, in 1990. He received the PhD degree in control science and engineering from the School of Automation, Southeast University, Nanjing, China, in 2020. He joined the College of Computer Science and Engineering, Shanghai Maritime University, Shanghai, China. His research interests include belief function theory, information fusion, and pattern recognition.



Tianyun Zhu received the BE degree in information and computational science from the Shanghai University of Electric Power, Shanghai, China, in 2023. He is currently working toward the MS degree with the College of Information Engineering, Shanghai Maritime University, Shanghai. His research focuses on theory and application of quantum mass function.

Xinde Li (Senior Member, IEEE) received the PhD degree in control theory and control engineering from the Department of Control Science and Engineering, Huazhong University of Science and Technology,Wuhan, China, in 2007. He joined the School of Automation, Southeast University, Nanjing, China, where he is currently a professor and PhD supervisor. His research interests include information fusion, object recognition, computer vision, intelligent robot, and human–robot interaction.

Jean Dezert was born in L'Hay les Roses, France, in 1962. He received the PhD degree from University Paris XI, Orsay, France, in 1990. Since 1993, he has been a senior research scientist with Information Modeling and Processing Department, ONERA, Palaiseau, France. His research interests focus on belief function theory, particularly for DSmT, which he has developed with Prof. Smarandache.

Rigui Zhou received the PhD degree from the Department of Computer Science and Technology, Nanjing University of Aeronautics and Astronauts, China, in 2007. From 2008 to 2010, he was a post doctoral fellow with Tsinghua University, China. He is currently a professor with the College of Information Engineering, Shanghai Maritime University, China. His research interests include quantum image processing, quantum reversible logic, and quantum genetic algorithm.

Changming Zhu received the PhD degree in computer science from the East China University of Science and Technology, Shanghai, China, in 2015. He is currently an associate professor with the College of Computer Science and Engineering, Shanghai Maritime University, Shanghai. His research interests include pattern recognition and machine learning.

Lei Cao received the PhD degree in computer software and theory from Tongji University, China, in 2016. He was a lecturer with Shanghai Maritime University, China. He was also a visiting researcher with the Institute of Medical Psychology and Behavioral Neurobiology, Tuebingen University in 2013. He has authored or co-authored about 20 journal and conference papers. His research interests include EEG signal processing, brain–computer interface, machine learning, and medical image analysis.

Shuzhi Sam Ge (Fellow, IEEE) received the BSc degree from the Beijing University of Aeronautics and Astronautics, Beijing, China, in 1986, and the PhD degree from the Imperial College of Science, Technology and Medicine, University of London, London, U.K., in 1993. He is currently the founding director with Robotics Institute and with the Institute of Intelligent Systems and Information Technology, University of Electronic Science and Technology of China, Chengdu, China. He is also the founding director with Social Robotics Laboratory, Interactive Digital Media Institute, National University of Singapore, Singapore, where he is a professor with the Department of Electrical and Computer Engineering. He is the editor-in-chief for *International Journal of Social Robotics*. He was an associate editor for a number of flagship journals. He was also the vice-president with Technical Activities from 2009 to 2010, Membership Activities from 2011 to 2012, and the IEEE Control Systems Society.